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CONCURRENT VALIDATION OF AN ARTIFICIAL INTELLIGENCE-BASED MOTION CAPTURE SYSTEM USING A COMMON FUNCTIONAL MOVEMENT SCREEN EXERCISE

By

Mitchell Nienhuis

A DISSERTATION

Submitted to the Faculty
of the University of Miami
in partial fulfillment of the requirements for
the degree of Doctor of Philosophy

Coral Gables, Florida

May 2024

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Concurrent Validation of an Artificial
Intelligence-Based Motion Capture System
Using a Common Functional Movement Screen Exercise

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The ability to accurately measure and quantify human motion is essential in both sport performance and musculoskeletal injury rehabilitation. By accurately quantifying human movement, optimization of biomechanics and muscle recruitment patterns can occur, while the identification and correction of contraindicated patterns can reduce the risk of injury or accelerate the rehabilitation process. Electromyography (EMG) is a common tool that is used to measure muscle activity and can be used to assess movement quality. However, due to limitations present when EMG is used in isolation, biomechanical analysis is often used concurrently to quantify movement. At present, marker-based optical motion capture systems are the gold standard in motion analysis; but they are not always ideal due to their high cost, need for dedicated space, and the expertise required for proper subject preparation. Conversely, markerless motion capture systems, are claimed by their manufacturers to be more cost effective and user friendly. Currently, the forward lunge is a popular exercise in both rehabilitative and human performance programs due to its unilateral nature, which has been shown to increase prime mover muscular strength, stabilizer muscle activity, and range of motion. Although the forward lunge is a functional exercise, due to the high quadriceps to hamstrings activation ratio (Q:H), physical therapists may avoid implementing it as a strategy during lower body

musculoskeletal rehabilitation protocols, particularly in anterior cruciate ligament (ACL) rehabilitation. The backwards lunge is a common alternative to the forward lunge. Therefore, the primary purpose of this dissertation was to compare the electromyographical and kinematic differences between the forward and backward lunge. A secondary purpose was to assess the validity of a markerless motion capture system in measuring joint angles of these lunge variations by comparing it to the gold standard optical system. The results of this dissertation showed that (Q:H) was significantly higher in the forward lunge than the backward lunge. Additionally, kinematic data showed a significantly greater knee angle in the forward lunge. For the markerless system, moderate to excellent agreement was shown between systems for both lunges. This study shows that different variations of the lunge exercise result in different joint angles, which may impact muscle activity. These results may be useful when selecting exercises during a rehabilitation plan. Furthermore, a markerless system may be a useful tool in quantifying movement in clinical settings once system accuracy improves.

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CHAPTER 1: INTRODUCTION

The human body is an intricate system comprised of several subsystems that work together to produce locomotion. When perfectly synchronized, the intricately wired central nervous system can coordinate complex neural patterns being sent to the muscle fibers that move the axial and appendicular skeleton producing the desired movement. Concurrently, sensory complexes in the periphery detect these movements and can transmit their data back to the central nervous system, where they can be used to make necessary adjustments to the descending signals. However, there are often flaws in this intricate network. The flaws may be as simple as suboptimal patterning when a new physical task or technique is learned, or as complex as a physical disruption of communication due to serious musculoskeletal injury. In either case, it is imperative to be able to identify these undesirable patterns to reduce the risk of injury.

Injuries can differ in both severity and cost. Research suggests that an injury as minor as an ankle sprain can cost over $500 per visit when expenses for equipment and personnel are considered, while more extensive injuries, like anterior cruciate ligament (ACL) injuries, can cost substantially more (Verhagen, van Tulder, van der Beek, Bouter, & van Mechelen, 2005; Vincent et al., 2022). In high stakes environments like collegiate or professional sports, injuries can mean missed playing time and millions of dollars in lost revenue if the result is a missed opportunity to compete in a national tournament or championship (Krautmann & Ciecka, 2009). With the high costs associated with repair

and rehabilitation of existing injuries, the ability to identify and prevent injuries before they occur should be of paramount importance.

For example, ACL injuries occur in as many as 1 in 29 female and 1 in 50 male athletes (Montalvo et al., 2019). These injuries can result in months of lost playing time due to surgery and rehabilitation (Waldron, Brown, Calderon, & Feldman, 2022). While some ACL injuries occur due to a collision with another athlete, it has been estimated that approximately 75% of injuries occur without contact (Boden, Dean, Feagin, & Garrett, 2000). It has been determined that in noncontact injuries, risk factors include multiplanar loading of the knee and imbalanced coordination of quadriceps and hamstrings contractions (Ahmad et al., 2006; Shimokochi & Shultz, 2008). Biomechanical studies have also shown that weakened gluteal muscles may predispose athletes to falling into knee valgus when performing a unilateral or single stance movement like cutting (Alentorn-Geli et al., 2009; Delp, Hess, Hungerford, & Jones, 1999). Therefore, an effective prevention strategy should include exercise selections that are designed to create a more balanced quadriceps to hamstrings activation ratio, as well as employing unilateral movements to strengthen the gluteals' ability to prevent femoral anteversion.

Optical motion capture systems are currently the gold standard for tracking and quantification of human movement (Van der Kruk & Reijne, 2018). These systems are used in the entertainment, clinical, and sports industries for a variety of purposes (Bini, Diefenthaeler, & Mota, 2010; Bregler, 2007; Eichelberger et al., 2016; Sutherland, 2002;

Wang, Kozak, Wan, Gomez-Levi, & Strumolo, 2013). Provided that all equipment is configured and calibrated properly, and a capture space is sufficient and free of obstacles, the degree of accuracy and reliability for the data capture session is very high (Eichelberger et al., 2016). Indeed, this technology is being implemented into training and rehabilitation protocols to identify movements that may place an individual at an increased risk of injury or re-injury. The Landing Error Scoring System, for example, is a field assessment that attempts to quantify landing mechanics and the injury risk. For example, Padua et. al. were able to utilize motion capture to validate the test and its ability to differentiate optimal vs sub-optimal or contraindicated landing mechanics (Padua et al., 2009).

Markerless motion capture systems, while not a recent development, have received increased attention in recent years as an alternative means of human movement tracking in clinical and sport settings, due to their enhanced portability, utility, and cost effectiveness. In our laboratory, we have validated the KinaTrax markerless motion capture system (KinaTrax Inc., Boca Raton, FL) in clinical populations and have showed it to be a high quality alternative to the gold standard marker-based system (Ripic, Nienhuis, et al., 2023; Ripic, Signorile, et al., 2023; Ripic, Signorile, Kuenze, & Eltoukhy, 2022). Given these results, validation research could be expanded into movement screens and prescriptive exercises to identify movement patterns or variations that may lead to injury.

Movement screens have gained in popularity as a field-based alternative to motion capture analysis for assessing an athlete's movement quality and stratifying their risk of musculoskeletal injury (Kraus, Schütz, Taylor, & Doyscher, 2014). Unlike traditional motion capture systems, which can measure joint angles and velocities, movement screens require an experienced test administrator or technician to monitor an athlete as they progress through a series of movements and assign a score based on the tester's perceived quality of that movement (Minick et al., 2010). Indeed, research has shown that these methods of categorizing movement quality can be reliable, provided the test administrator has a high level of training and expertise (Gribble, Brigle, Pietrosimone, Pfile, & Webster, 2013; Minick et al., 2010). However, it has also been shown that movement screens may not have a high degree of validity in predicting injury or identifying movement dysfunction (Warren, Lininger, Chimera, & Smith, 2018). A meta-analysis conducted by Moran and colleagues found that a functional movement screening test battery was not effective as an injury prediction tool (Moran, Schneiders, Mason, & Sullivan, 2017) because these screening tests take scores of individual movements and add them together for a composite score. This creates the potential for errors and nonspecific results, which do not provide precise identification of musculoskeletal structures that are at an increased risk of injury. By isolating individual movement assessments, a more precise report could be created that could identify specific body segments that may be at risk of injury.

Indeed, recent work is shifting toward isolating individual movements in an attempt to better categorize movement proficiency and injury risk (Chek, 2000; Cook, 2003; Gadomski, Ratamess, & Cutrufello, 2018; Kritz, Cronin, & Hume, 2009). Several

of these screening exercises attempt to replicate, in whole or in part, the fundamental movement patterns that are performed in activities of daily living, sports, and strength training (Chek, 2000). One very common movement is the lunge. While considered a foundational movement, it is also commonly used in rehabilitation, as well as strength and conditioning settings (Boudreau et al., 2009; Keogh, 1999; Kritz et al., 2009; Marchetti et al., 2018; Nadzalan et al., 2021; Pincivero, Aldworth, Dickerson, Petry, & Shultz, 2000). As it is unilateral in nature, it can be useful for identifying weakness in hip and knee musculature, that can cause undesirable and contraindicated movement asymmetries between limbs (Kritz et al., 2009). However, biomechanical and electromyographical (EMG) studies have shown that the traditional lunge exercise can create significant stress to the patellofemoral joint and a high quadriceps-to-hamstrings activation ratio (Q:H), which leads to anterior shearing stress on the knee joint (Begalle, DiStefano, Blackburn, & Padua, 2012; Ekstrom, Donatelli, & Carp, 2007; Escamilla et al., 2022). This may be contraindicated for certain populations that need to minimize patellofemoral joint stress, like anterior cruciate ligament (ACL) rehabilitation patients. Therefore, an alternative to the traditional lunge should be investigated.

The backward lunge is also commonly used in training settings and is similar to the forward lunge in its positioning and ability to increase hip and knee muscular strength (Keogh, 1999; Krause et al., 2018; Kritz et al., 2009; McClellan & Bugg, 1999). What is less known, is whether performing the backward lunge produces a similar activation pattern of the hip and knee musculature and can therefore be used as a substitute for the traditional lunge during rehabilitation protocols or as a functional screening exercise. The

limited research that has investigated the backwards lunge has found that it produces less patellofemoral joint stress (Goulette, Griffith, Schiller, Rutherford, & Kernozek, 2021) and lowers quadriceps activation (Park, Huang, Song, & Lee, 2021) than the forward lunge. However, to our knowledge, no research has investigated differences in Q:H or kinematics at the hip and knee between the two exercises.

Therefore, this dissertation has three aims.

Aim 1

To compare the differences in peak and mean quadriceps and hamstring activity levels, and Q:H between the forward and backward lunges. **Hypothesis:** The forward lunge will produce a higher Q:H than the backward lunge.

Aim 2

To compare the kinematic differences between the forward and backward lunge at the hip and knee joint. **Hypothesis:** There will be no differences in joint angles at the hip and knee between lunge techniques.

Aim 3

To determine the validity of the kinematic data from the forward and backward lunge calculated using an eight-camera markerless system compared to a gold standard marker-based optical system. **Hypothesis:** The kinematics from the eight-camera markerless system will compare favorably with the kinematics from the gold standard optical system.

CHAPTER 2: LITERATURE REVIEW

2.1 Overview and History of Movement Analysis

Humans have been fascinated by their anatomy and physiology since ancient times, as evidenced by the paintings and portraits of men and women in different poses and activities (Barringer, 2014); however, the interest in the human body did not stop at mere paintings. Susruta, a physician in 600 B.C. India, was the first documented physician to prescribe moderate exercise to his patients (Susruta, 2008; C. M. Tipton, 2014). In the first half of the third century B.C., Greek scientists were documented to have performed systematic dissections of human cadavers (von Staden, 1992). Additionally, the ancient Greek civilizations popularized physical fitness and movement proficiency to optimize their health and performance. The Spartans prioritized fitness with the goal of becoming elite warriors and would often participate in strenuous exercise and combative athletics to hone their physical skills (Hooker, 1980; Mitchell, 1952; C. Tipton, 2008).

At the onset of the Renaissance, this Greek culture and knowledge migrated westward and influenced intellectuals in Europe. Leonardo da Vinci is famous as a pioneer in the field of anatomy, due to his extensive skill in human dissection and anatomical illustrations (Da Vinci, 2013). He classified muscles by their shape and successfully identified mechanical functions of the limbs and their possible ranges of motion around their joints (Clayton & Philo, 2010; Shephard & Shephard, 2015).

By the 1800's scientists had begun identifying and evaluating muscle function and had learned that it was caused "by the conduction of electricity" (Hale, 2008). By the 1950s, "modern" exercise physiologists had begun to intently research the human body, its response to physical stress, and how to optimize athletic performance (Banta, 1999; Hale, 2008).

In 1836, the Weber brothers quantified human locomotion using temporal and spatial parameters (Weber & Weber, 1836). By the 1870s the earliest kinematic studies on human gait were performed using still cameras (Whittle, 1996). The 1940s yielded the invention of videography, which allowed for the capture of human gait in motion (Inman, Ralston, & Todd, 1981). Finally, by the 1980s, measurement systems that utilized computers based on recordings from television cameras became available, improving the time required for calculations and the convenience of data collection (Whittle, 1982).

2.1 Current Methods for Motion Capture Analysis

Currently, the gold standard for capturing and analyzing human movement includes using multiple calibrated video-recording cameras and attaching markers to subjects that can be identified and tracked by optical motion capture software (Van der Kruk & Reijne, 2018). Markers placed on subjects can be classified as active or passive. In active markers, each marker contains a light-emitting diode (LED) that is detected by the camera system. Conversely, passive systems utilize reflective markers that are placed

on subjects and reflect infrared light that is emitted from the camera system (van Schaik & Dominici, 2020). For optical motion capture to work, markers need to be visible by a minimum of two cameras. The markers can then be tracked by the system and their positions can be estimated by triangulation between the cameras and a calibrated reference position. While both active and passive markers may be effective options, passive marker systems are preferred due to the lower cost of the markers, the lesser weight, and the elimination of the need to ensure all markers have adequate electrical power (van Schaik & Dominici, 2020). One limitation of current optical motion capture systems is the movement artifacts when markers are placed on soft tissue, which may cause skeletal movement and bone segment dimensions to be overestimated (Mundermann, Corazza, & Andriacchi, 2006). While techniques exist to minimize this error, they are either invasive, or confining to the point where fluid movement is restricted. Additionally, the systems' costs, space requirements, lack of portability, and the technical skill required for use, reduce their feasibility for most organizations.

2.2 Markerless Motion Capture Analysis

Markerless motion capture systems have significantly improved over the past two decades. Originating from the fields of computer vision and machine learning, markerless motion capture is now entering the field of human movement and is being implemented in clinical settings for the evaluation of musculoskeletal disease and dysfunction (Andriacchi & Alexander, 2000; Mundermann et al., 2006). Even though some limitations still exist in the system's capacity to accurately identify and represent human movement, results are promising and by incorporating artificial intelligence and machine

learning, the validity and reliability of these systems are increasing rapidly and they are being incorporated into more laboratories and sports facilities (Kanko et al., 2021; Kotsifaki, Whiteley, & Hansen, 2018; Ripic, Signorile, et al., 2023; Ripic et al., 2022). Although many markerless motion capture systems exist, the preferred systems contain multiple cameras that increase the probability of at least two cameras being able to capture each body segment (Armitano-Lago, Willoughby, & Kiefer, 2021). KinaTrax (KinaTrax Inc., Boca Raton, FL) is a multi-camera system that has been utilized primarily for baseball, but is expanding into use with more clinical populations (Dobos et al., 2022; Nebel, 2022; Ripic, Signorile, et al., 2023). Our laboratory is currently validating the viability of using KinaTrax software for assessing movement in multiple populations. In 2022, Ripic et al. showed that KinaTrax exhibited strong agreement with a marker-based system in determining gait speed, stride length, step length, cycle time, and step time in healthy young adults (Ripic et al., 2022). In 2023, work expanded to include healthy adults and older adults with Parkinson's disease, again showing excellent agreement in all spatial and temporal values (Ripic, Signorile, et al., 2023). Once spatiotemporal parameters were validated, a follow-up study was conducted in which kinematics were compared between KinaTrax and a marker-based system using healthy populations and adults with Parkinson's disease, which again showed excellent agreement between systems (Ripic, Nienhuis, et al., 2023). These results demonstrated that the KinaTrax markerless motion capture system can be an effective and useful tool to assess biomechanics in both healthy and clinical populations. Research should continue to examine the validity of this software for other clinical populations with the aim of

identifying pathological movements that may be indicative or predictive of musculoskeletal injury.

2.3 Electromyography

Electromyography (EMG) is a tool that is commonly used to assess muscle activity during movement. The history of EMG can be traced back to the 18th century when Luigi Galvani and colleagues accidentally discovered that the muscles of a dissected frog contracted when the nerve was touched by an electrically charged metal knife (Kazamel & Warren, 2017; McComas, 2011). In the 19th century, the experiments of Galvani were continued and expanded to confirm that muscle contraction was due to an electrical signal traveling down a nerve and not due to activation of the muscle by the "soul" (Brazier, 2012). These experiments also expanded to discover that muscles responded differently if they were stimulated in different locations, which eventually came to be referred to as muscle motor points (Duchenne, 1855; Remak, 1858).

By the 20th century, the interest in muscle activity began to shift from manual stimulation to measuring internal activation. The concept of the motor unit was established by Edgar Douglas Adrian and Detlef Bronk when they used a needle electrode connected to a loudspeaker to record the electrical activity of the muscle fibers that were innervated by a single nerve fiber (Adrian & Bronk, 1929). Surface EMG (SEMG) appeared in the 1940s and gained popularity due to its relationship to research

on dynamic movement, particularly when assessing muscle activity in the shoulder and the torso (Cram, 2003).

Currently, the EMG signal is considered a manifestation on the muscle of the electrical activity sent along the alpha motor neurons, which are activated by the spinal cord as a result of either conscious or unconscious stimulation by the brain (Kleissen, Buurke, Harlaar, & Zilvold, 1998). These alpha motor neurons transmit action potentials due to changes in electrical charge at the cell membrane, which are caused by the flux of sodium and potassium ions. As the action potential reaches the neuromuscular junction, the electrical signal is transferred to the muscle. As these electrical signals pass by the EMG electrodes, the voltage changes are recorded and sent to a differential amplifier, which can eliminate any transient noise and amplify only the signal caused by the electrical activity of the muscle. The combination of multiple motor unit activation and rate coding results in the raw EMG signal that is recorded.

Electrodes that are placed over muscle bellies can detect the electrical activity within the muscle and can be used to gauge overall activity when performing a particular action. Amplitude comparisons between muscles and between movements can be used to identify which muscles are preferentially recruited. For example, Slater and colleagues compared muscle activities between two different types of squats and found that when the knees were misaligned with the feet, biceps femoris and gastrocnemius activity levels were elevated when compared to a squat in which the knees and toes were more

optimally aligned (Slater & Hart, 2017). This ability to compare diverging activation patterns may allow EMG to be used in a clinical setting to identify potential risk factors for injury. In a systematic review of the literature done by Barton et. al., delays and reduced duration of activity in gluteus medius were shown to be related to patellofemoral joint pain (Barton, Lack, Malliaras, & Morrissey, 2013). It should be noted, however, that the use of EMG in movement evaluation has limitations in both technique and processing, including amplitude cancellation, crosstalk by adjacent muscles, and time scale normalization (Hug, 2011). One way to address some of these limitations is by the concurrent assessment of biomechanical factors, such as joint kinematics, which have been found to be a significant determining factor in muscular activation and coordination (Campanini, Disselhorst-Klug, Rymer, & Merletti, 2020; Hug, 2011; von Werder & Disselhorst-Klug, 2016). By incorporating EMG into clinical movement evaluations, there is the potential to identify possible mechanisms that may increase or decrease the stresses placed on a joint.

2.4 ACL Injury and Mechanisms of Rehabilitation

Anterior cruciate ligament injuries are some of the most common knee injuries treated by physical therapists (Bollen, 2000). During the strengthening component of the rehabilitation process, one of the primary goals of the therapist is to help the client regain quadriceps strength and control of the knee joint (Cavanaugh & Powers, 2017). Traditionally, open kinetic chain (OKC) exercises were avoided due to the increased anterior shear stress on the ACL graft. The seated leg extension exercise, for example, has been identified as a contraindicated exercise during early rehabilitation because of the

peak anterior shear forces caused by the contraction of the quadriceps and concurrent lack of hamstrings activity (Wilk, Arrigo, Bagwell, & Finck, 2021).

Additionally, the use of multi-joint exercises can increase the coactivation of the hamstrings group, which assists the ACL in preventing anterior displacement of the tibia (Draganich, Jaeger, & Kralj, 1989; Hefzy, Al Khazim, & Harrison, 1997). These exercises can be useful for reducing the overall stress on the ACL while working to restore quadriceps strength and control. Furthermore, the coactivation of the quadriceps and hamstrings may be a more important factor affecting reinjury of the ACL during rehabilitation than whether an OKC or closed kinetic chain (CKC) exercise is employed. As was previously stated, the risk associated with the seated knee extension exercise was due to anterior shear forces, which are maximized from 0-40 degrees of knee extension, and the recruitment of only the quadriceps during the movement (Wilk et al., 1996). Since quadriceps activation creates anterior shearing forces and hamstrings activation creates posterior shearing forces, logically, the OKC knee extension would place the ACL at greatest risk (Beynnon et al., 1995). However, it was noted that the leg press exercise, which is also generally classified as an OKC activity, did not produce any significant anterior shearing forces (Wilk et al., 1996). Therefore, the quadriceps to hamstrings ratio (Q:H), which is calculated as the average quadriceps activation divided by the average hamstrings activation during a particular movement, may be a better way to assess potential injury risk to the ACL during that exercise (Begalle et al., 2012).

2.5 Unilateral Exercise

Unilateral exercises are a commonly utilized tool for clinicians (Moore, Semciw, & Pizzari, 2020). Unlike the more traditional exercises like the barbell squat or bench press, unilateral exercises rely on a single limb to accomplish the prescribed movement. This allows practitioners to focus on one limb for rehabilitation purposes and prevents any compensatory movement or favoring of the unaffected limb during training (Baumgart, Schubert, Hoppe, Gokeler, & Freiwald, 2017). Unilateral exercises may also be preferred due to their ability to introduce patients to an unstable environment, which is a key component of improving trunk stability and balance (Behm, Leonard, Young, Bonsey, & MacKinnon, 2005). Additionally, performing unilateral exercises has been shown to increase motor unit recruitment to a greater degree than bilateral exercises (Sale, 1988).

2.6 The Traditional Lunge

One such unilateral exercise is the lunge. While some disagreement may exist regarding its inclusion as a true unilateral exercise, previous research has shown that the lead leg does accomplish most of the work needed to complete the movement (Fountaine, 2018). The lunge is a CKC movement that is traditionally characterized by the participant starting in an upright standing position and then taking a moderately large step forward so that the thigh of the lead leg is parallel to the ground (Keogh, 1999). Most practitioners coach their clients to control their descent so that they stop just before the trailing knee contacts the ground, to prevent injury. While some variations may include a larger step

into a broad lunge, most commonly, the client will utilize a moderate step that results in the back knee ending at approximately a 90-degree angle (Distefano, Blackburn, Marshall, & Padua, 2009). The exercise ends when the client uses the lead leg to forcefully propel themself back to the starting position. This step-and-return is the fundamental component that differentiates the lunge from a similar movement like the split squat (Keogh, 1999; Schütz et al., 2014). The primary muscles responsible for this sagittal plane movement are the vastus lateralis, vastus medialis, biceps femoris, gluteus maximus, and gluteus medius (Boudreau et al., 2009; Marchetti et al., 2018; Pincivero et al., 2000). Although research has shown that the lunge exercise is important in clinical and general strength training, as it can isolate and identify strength asymmetries between limbs (Kritz et al., 2009; Nadzalan et al., 2021); due to its high Q:H of almost 10:1, an alternative exercise offering the same benefits with a reduced Q:H would be more viable (Begalle et al., 2012; Ekstrom et al., 2007).

2.7 The Backward Lunge

Although the forward lunge may be contraindicated for ACL rehab patients, other variations of the lunge may provide a safer alternative. Lunge variations have been shown to change muscle activation and joint loading, while positional changes of the trunk and the step length can redistribute the total work on the body. Furthermore, performing a forward lunge with a forward lean has been shown to increase the activity of the biceps femoris when compared to a lunge performed with an upright trunk, although it does not appear to reduce the Q:H (Farrokhi et al., 2008). Step length has also been shown to alter the demands on the knee joint. Escamilla and colleagues showed that lunging with a short

step length causes anterior tibial translation over the toes and results in joint forces almost two times higher than a lunge with a longer step (Escamilla et al., 2022). Variations of the lunging motion, like the lateral and reverse lunges, may also impact the Q:H (Begalle et al., 2012; Keogh, 1999; McClellan & Bugg, 1999). Lateral lunges were found to place greater demands on the knee and ankle when compared to the traditional forward lunge (Riemann, Congleton, Ward, & Davies, 2013). Additionally, lunging with the back foot suspended resulted in almost twice as much activation in the hamstrings and gluteus maximus and medius when compared to the traditional forward lunge (Krause et al., 2018); but, without significant differences in rectus femoris activation between those variations, the Q:H was significantly improved. However, an examination of the methodology used in this study indicated that to accomplish the suspended lunge, the subjects were required to perform a variation of a backward lunge. This generates a question of whether the backward lunge can reduce the anterior shearing forces on the knee while producing an optimal activation ratio between the quadriceps and the hamstrings.

Recent work examined this theory. One recent study, which included a sample of young adult females, showed higher patellofemoral joint loading during the forward lunge when compared to the backward lunge; however, this study did not examine the differences in Q:H using electromyography (Goulette et al., 2021). In the only other study that examined differences between the forward and backwards lunge, Park et al. compared muscle activity and vertical ground reaction forces between the forward and backward lunges in a sample of young adult males. They found that the vastus medialis

oblique activation (VMO) and peak ground reaction force were higher during performance of the forward lunge (Park et al., 2021); however, there was no quantification of Q:H. This research indicates that there may be biomechanical and electromyographical differences between these two variations; however, given the limited research on the backward lunge, these differences should be further investigated. More detailed review of the literature, as well as a thorough review of other literature that includes visual representations of the lunges (Goulette et al., 2021; Keogh, 1999; Krause et al., 2018; McClellan & Bugg, 1999; Park et al., 2021), it appears that the beginning and the point of eccentric to concentric transition, or the "bottom," are very similar. Research should be conducted to investigate the differences that exist between these two lunge variations. Additionally, the ability to verify these differences using different types of markerless motion capture analysis software may improve the diagnostic abilities of rehabilitation centers as these user-friendly become more available.

CHAPTER 3: METHODS

3.1 Subjects

A total of 34 healthy subjects (men = 21, women = 13) were recruited from the University of Miami. Subjects ranged from 18-30 years of age (mean: 21.62 years, SD: 3.14), were recreationally active, and were free of any musculoskeletal injuries for the previous 6 months prior to testing. This study was approved by the University's Institutional Review Board and the testing protocol, benefits, and risks of the study were explained to the participants prior to them reading and signing a university-approved informed consent.

3.2 Experimental Design

The study employed a cross-sectional design, where individual subjects were brought into the laboratory for a single 90-minute testing period. Following the informed consent, subjects had their height and mass measured in centimeters and kilograms, respectively. Subjects were instructed to arrive in tight-fitting or compression clothing. If subjects' clothing was likely to cause error in the marker-based system, they were asked to change into provided spandex clothing. Next, subjects' dominant leg length was identified based on their answer to the question, "if you were to kick a soccer ball as hard as you can, which leg would you kick it with?" The dominant leg was measured according to established best practices to identify lunge step length (Sabharwal & Kumar, 2008). Subjects then had surface electromyography sensors (Trigno Avanti Sensor,

Delsys, Natick, MA, USA) placed on the vastus medialis, vastus lateralis, rectus femoris, biceps femoris, and medial hamstrings on the dominant leg in accordance with the guidelines published by Criswell and colleagues (Criswell, 2010). Reflective markers were placed on anatomical landmarks according to a modified plug-in gait model by Vicon (Vicon LLS, Coral Gables, FL, USA) with the addition of an extra marker on each body segment to ensure a minimum of 3 markers were visible by the camera system at all times (see Figure 1).

3.3 Electromyography

A bipolar surface electrode configuration with an interelectrode distance of 1 centimeter was used to maximize the reception area. Research shows that this distance is effective in reducing the risk for crosstalk in most muscle groups (Criswell, 2010). Sensor sites were identified and marked with white pencil using parallel lines at a distance of ~ 2 inches. These marked spots were prepared for the EMG sensors by shaving the area to remove body hair, abrading with sandpaper to remove loose skin cells, and cleansing with an alcohol pad to remove any residual debris or oils thereby reducing the impedance at the skin surface. Once sensors were in place, data were collected using EMGWorks Acquisition software (Delsys, Natick, MA, USA). The gain was set at 2,000 with bandpass filtering set between 20-450 Hz. Signals were sampled at a frequency of 2,000 Hz, digitized using a 16-bit A/D converter, and stored on a laboratory computer.

3.4 Motion Capture

A total of 54 reflective markers were placed on major body landmarks according to a modified plug-in gait model by Vicon (Vicon LLC, Coral Gables, FL, USA) with the addition of an extra marker on each body segment to ensure a minimum of three markers are visible by the camera system (see Figure 1). These additional markers were placed bilaterally on the medial femoral condyles, medial malleoli, antero-inferior thighs, antero-inferior shanks, the iliac crests, and on the proximal fifth metatarsophalangeal joints. A 10-camera motion capture system was used to track the markers during the execution of the lunges (Qualysis AB, Gothenborg, Sweden). Calibration of the system was in accordance with the directions in the user manual. Reconstructed static trials were visually inspected to ensure all markers were present. Frame capture rate was set at 100 frames per second (fps). An eight-camera markerless system (KinaTrax Inc., Boca Raton, FL) was used concurrently to capture lunge trials at 100 fps. A background video of the motion capture space was taken prior to recording dynamic trials. This background was taken for each testing day to correct any camera drift (i.e., misalignment) between the testing days and the latest system calibration background. The negative background image from the testing day was aligned manually with the calibration background using the force plate area as reference. This ensured that small changes in camera orientation were accounted for in the 3D reconstruction of the markerless key points, which otherwise would be a source of inaccuracy.

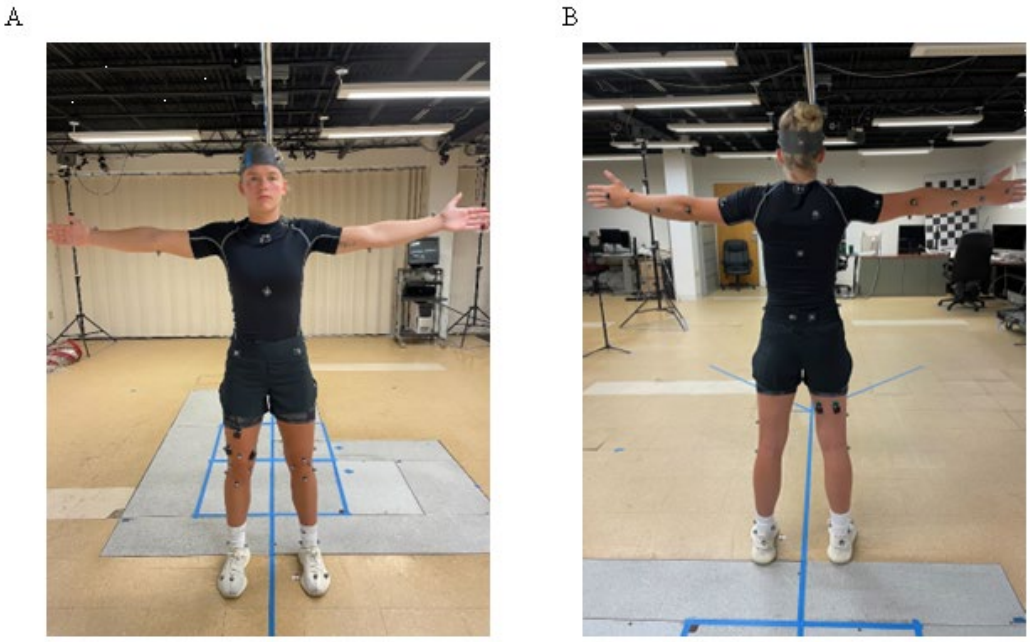


Figure 1: Reflective marker and surface EMG placement.

3.5 Testing Procedure

Calibration of the marker-based system with the subject incorporated both a static and dynamic calibration protocol. After all systems were calibrated, subjects were allowed to perform a familiarization set of 3-5 forward and backward lunges. For the familiarization set, two force plates were oriented side-by-side and the subjects stood with one foot on each plate. Subjects were then asked to begin their familiarization set by performing 3-5 lunges forward to a taped distance equaling 75% of dominant leg length. After completion of the forward lunges, subjects then performed the same number of lunges in the reverse direction by stepping off the force plate with their nondominant leg and lunging backward to a taped mark equal to 75% of the length of the dominant leg. These data were used for validation of event determination (see Section 3.6.4 for details).

The order in which subjects performed the forward or backward lunge first was randomly assigned. For the forward lunge trial, the subject started at a taped mark on the ground that was a distance of 75% of the dominant leg length from a designated target (see Figure 2A), in accordance with previous research (Goulette et al., 2021). Subjects performed five individual lunges by stepping toward the target point with their dominant leg. For the backwards lunge, the subjects started at the designated target and step back to the taped mark with the nondominant leg. Subjects were asked to keep the heel of the dominant leg on the ground at all times when lunging to prevent excessive patellofemoral stress (see Figure 3) (Chandler, McMillan, Kibler, & Richards, 2002; Zellmer, Kernozek, Gheidi, Hove, & Torry, 2019). Subjects were also asked to maintain an upright torso position for all trials to reduce any differences in muscle activation that could be attributed to different torso angles (Farrokhi et al., 2008).

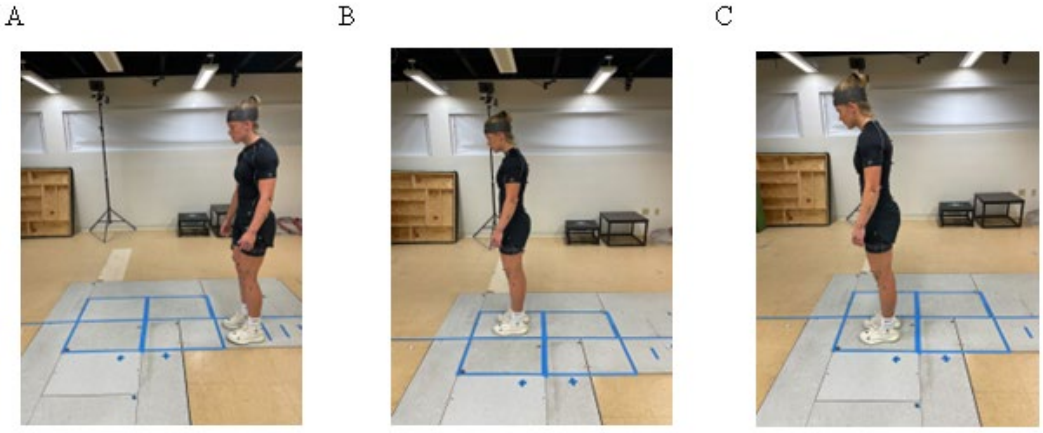


Figure 2: Starting positions for the (A) forward lunge, (B) backward lunge before trial capture, and (C) backward lunge.



Figure 3: Bottom position of the lunge exercise.

3.6 Data Analysis

3.6.1 Electromyography

Data were collected using EMGWorks Acquisition software and exported to EMGWorks Analysis for processing (Delsys, Natick, MA, USA). EMG was not normalized, as all data from each subject were collected on the same day. Data were rectified using root mean square:

$$RMS = \left(\frac{1}{S}\sum_{1}^{S} f^2(s)\right)^{1/2}$$

After the data were rectified, they were exported to Microsoft Excel for further analysis (Microsoft, Redmond, WA, USA). Peak activation was determined by finding the highest recorded value and then taking a 200-millisecond window around the value and calculating the average. Averages from each muscle from each repetition were calculated in this way. The three quadriceps muscles and two hamstrings muscles were averaged to calculate a mean quadriceps and hamstrings amplitude. The middle three trials for each lunge variation were averaged to calculate a mean peak activation and then divided to calculate the quadriceps to hamstrings activation ratio.

3.6.2 Marker-Based 3DMA

Marker-based files were transferred from the collection system to an offline processing station. Trials were reviewed to ensure that all markers and identified body segments were present during trials and the capture segments were cropped to include only the lunge trial. Missing markers were gap-filled using a relational function based on three other markers present on the same body segment. In cases where three markers were not present, a single marker on the same segment was used to gap-fill; and in cases where less than 10 frames were missing, gaps were filled using cubic spline fitting. Processed files were then imported into Visual 3D (C-Motion Inc., Germantown, MD) for skeletal modeling and lunge event determination.

3.6.3 Markerless 3DMA

After capture, files were compiled and processed using the *HumanVerson3* model from KinaTrax (see Figure 4). *HumanVersion3* is a full body model that contains 40 landmarks with newly trained points added bilaterally on the lower body containing the calcaneus, first metatarsal head, fifth metatarsal head, hallux head, and the patella center. A more comprehensive explanation of the markerless processing procedures used in this laboratory has been outlined previously (Ripic et al., 2022).

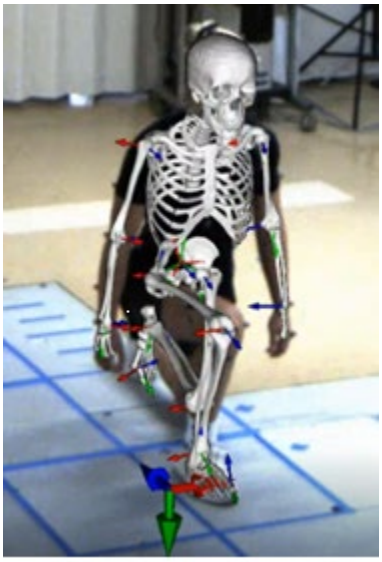


Figure 4: KinaTrax skeletal modeling using HumanVersion 3.

After local processing on the KinaTrax system, data were imported into Visual 3D for skeletal modeling and lunge event determination.

3.6.4 Lunge Event Determination

Currently, the different events of the lunge are determined using force plates, specifically the onset of force on the plate as the initiation of the lunge and the cessation of force on the plate as the termination of the lunge (Goulette et al., 2021). In gait studies, force plate data are considered the gold standard (Zahradka et al., 2020). However, for the lunge, there are several limitations associated with this method. First, by including only data between initial contact (IC) and the terminal contact (TC), any kinematic and electromyographical data during the flight phases of the lunge, where the subject initiates the lunge from a standing position and returns to the standing position after TC, were eliminated. Second, for the backward lunge, there are no data for the onset or cessation of force, since this would require the subject to stand on the force plate for the entire trial. Finally, since the markerless system is not connected to the force plate, using kinetics to identify movement segments would be impossible. Based on these limitations, only kinematic events were used to identify lunge segments.

There is no current literature that identifies lunge segments based on kinematic events; however, there are several methods to determine gait events using only kinematic events. Zahradka and colleagues compared these methods and found that tracking velocity differences would correlate best with actual gait events (Zahradka et al., 2020). Based on these data, we elected to identify the lunge events as follows: the start of the lunge occurred when the velocity of the heel of the nondominant leg (i.e. the leg that would always be the trail, or back leg) relative to the pelvis exceeded -0.2 m/s in the anteroposterior direction, the bottom of the lunge was identified as the minimum value of

the pelvis in the vertical direction, and the end of the lunge was defined as the threshold crossing of the anteroposterior heel velocity relative to the pelvis at 0.2 m/s. This was done in accordance with previous work done in this laboratory, which used similar methods to identify gait events using only kinematic data of the pelvis and foot segments (Ripic et al., 2022).

Because of the low velocity of the movement and because only the discreet variables of knee and hip angle were included in the final analyses of the lunges and comparisons between the two systems, all kinematic data were filtered at 1 Hz to eliminate any spikes or artifacts. These data were compared with another dataset that was collected during the warmup lunges, in which the subjects stood on two adjacent force plates (Type 9286AA; Kistler Instrument AG, Winterhur, Switzerland). Validity of the kinematic events were compared to events as defined by force plate data. Thresholds of vertical ground reaction force were set at 20 N to determine the beginning and end of the lunge, which is similar to the threshold that is usually used during gait event determination (Ripic et al., 2022; Zahradka et al., 2020; Zeni, Richards, & Higginson, 2008).

3.7 Statistical Analysis

For the comparison of joint angles and Q:H between the two lunge variations, a 2-way repeated measures ANOVA was used with groups being identified as (1) method of lunge (forward or backward) and (2) grouping or counter effect (forward first or

backward first). For the validation of the markerless system at identifying the hip and knee joint angles at the bottom of the lunges, interclass correlation coefficients (ICCs) were used. ICC values less than 0.5 indicated poor validity, values of 0.5-0.75 indicated moderate validity, values of 0.75-0.9 indicated good validity, and values greater than 0.9 indicated excellent validity (Koo & Li, 2016). If ICC values were calculated to be less than 0.9, a 2-way repeated measures ANOVA was used to determine if the method of lunge or motion capture system had an interaction effect. The groups were identified as (1) method of lunge (forward or backward) and (2) method of measure for joint angle (marker-based vs. markerless. For all repeated measures ANOVA tests, eta squared values were interpreted as being small (<.01), medium (.06), or large (>.14) effects, according to recommendations published by Cohen and Cohen (Cohen, Cohen, West, & Aiken, 2013).

CHAPTER 4: RESULTS

4.1 Subjects

A flowchart of participants through the project is shown in Figure 5. There were no significant differences between groups for any baseline demographics. Of the 34 subjects recruited, complete data from 32 subjects were used for the EMG study, 29 subjects for the kinematics study, and 23 subjects for the validation study.

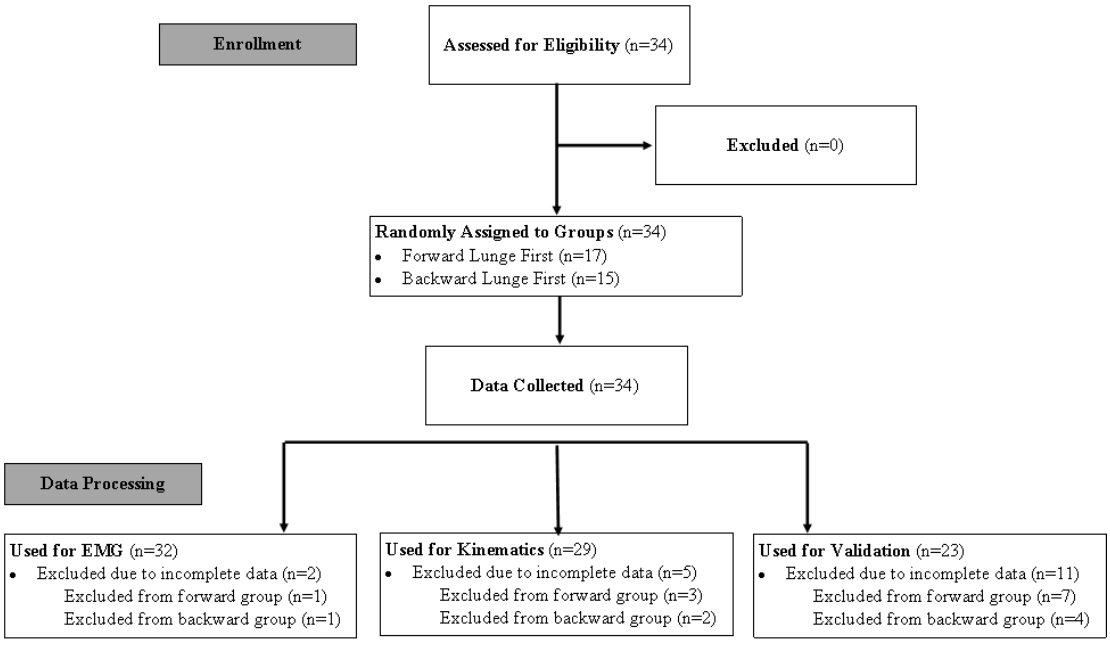


Figure 5: A consort chart of the flow of subjects through the study.

4.2 Electromyography

Mean values for peak EMG amplitude for each muscle group are presented in Table 1 and Table 2 for forward and backward lunges, respectively.

Table 1: Descriptive statistics for muscles for the forward lunge by group.

Group		N	Minimum	Maximum	Mean	Std. Deviation
Bkwd	Rectus Femoris	15	0.072	0.285	0.125	0.057
	Vastus Lateralis	15	0.072	0.353	0.203	0.087
	Vastus Medialis	15	0.216	0.588	0.401	0.119
	Quad	15	0.154	0.367	0.243	0.067
	Biceps Femoris	15	0.016	0.057	0.031	0.011
	Semitendinosus/ Semimembranosus	15	0.015	0.114	0.043	0.024
	Ham	15	0.018	0.086	0.037	0.017
	Q:H	15	2.932	15.862	7.347	3.053
	Valid N (listwise)	15				
Fwd	Rectus Femoris	17	0.057	0.482	0.134	0.102
	Vastus Lateralis	17	0.078	0.420	0.213	0.105
	Vastus Medialis	17	0.183	1.584	0.469	0.320
	Quad	17	0.144	0.756	0.272	0.143
	Biceps Femoris	17	0.018	0.211	0.054	0.051
	Semitendinosus/ Semimembranosus	17	0.020	0.134	0.049	0.028
	Ham	17	0.020	0.142	0.051	0.036
	Q:H	17	1.142	11.157	6.380	2.551
	Valid N (listwise)	17				

Data presented in millivolts. Fwd: group that performed the forward lunge first, Bkwd: group that performed the backward lunge first, Quad: mean of all quadriceps muscles, Ham: mean of all hamstrings muscles.

Table 2: Descriptive statistics for muscles for the backward lunge by group.

Group		N	Minimum	Maximum	Mean	Std. Deviation
Bkwd	Rectus Femoris	15	0.041	0.127	0.075	0.025
	Vastus Lateralis	15	0.057	0.286	0.159	0.063
	Vastus Medialis	15	0.116	0.707	0.327	0.151
	Quad	15	0.121	0.324	0.187	0.065
	Biceps Femoris	15	0.018	0.088	0.036	0.018
	Semitendinosus/ Semimembranosus	15	0.015	0.748	0.084	0.185
	Ham	15	0.019	0.418	0.060	0.100
	Q:H	15	0.567	14.728	5.472	3.143
	Valid N (listwise)	15				
Fwd	Rectus Femoris	17	0.042	0.375	0.108	0.076
	Vastus Lateralis	17	0.068	0.537	0.171	0.112
	Vastus Medialis	17	0.116	1.503	0.398	0.311
	Quad	17	0.087	0.679	0.226	0.131
	Biceps Femoris	17	0.018	0.106	0.040	0.022
	Semitendinosus/ Semimembranosus	17	0.018	0.115	0.044	0.027
	Ham	17	0.018	0.111	0.042	0.023
	Q:H	17	2.233	8.918	5.773	1.948
	Valid N (listwise)	17				

Data presented in millivolts. Fwd: group that performed the forward lunge first, Bkwd: group that performed the backward lunge first, Quad: mean of all quadriceps muscles, Ham: mean of all hamstrings muscles.

4.2.1 Comparison of Quadriceps EMG Activity Between Lunge Type

Mean quadriceps activity for the forward lunge was calculated to be 0.259 mV (SD = 0.113) and was calculated to be 0.208 mV (SD = 0.105) for the backwards lunge. Results indicated that a significant main effect was found for quadriceps activity ($F(1,30) = 37.75$, $p < 0.01$, $\eta^2 = .557$) but no interaction effect existed (See Table 3). Eta squared value of 0.557 indicated a strong effect.

Table 3: Repeated measures ANOVA[a] for quadriceps electrical activity between forward and backward lunge by group.

Effect	F	Hypothesis df	Error df	Sig.	Partial Eta Squared
Quad Activity	37.752[b]	1.000	30.000	< 0.001	0.557

a. Design: Intercept + Group
b. Exact
df: degrees of freedom.

4.2.2 Comparison of Mean Hamstring EMG Activity Between Lunge Type

Mean hamstrings activity for the forward lunge was calculated to be 0.045 mV (SD = 0.029) and was calculated to be 0.051 mV (SD = 0.070) for the backward lunge. Results indicated that no significant main effect was found for hamstrings activity and no interaction effect existed.

4.2.3 Comparison of Q:H Activity Between Lunge Type

Side by side comparisons of mean Q:H values for forward and backward lunges for each group are presented in Table 4.

Table 4: Descriptive statistics for quadriceps to hamstrings coactivation ratios by group.

Group		Mean	Std. Deviation	N
Forward_Q:H	Bkwd	7.347	3.053	15
	Fwd	6.380	2.551	17
	Total	6.833	2.794	32
Backward_Q:H	Bkwd	5.472	3.143	15
	Fwd	5.773	1.948	17
	Total	5.632	2.539	32

Q:H: quadriceps to hamstrings activation ratio, Bkwd: group that performed the backward lunge first, Fwd, group that performed the forward lunge first.

Results from the 2 by 2 repeated measures ANOVA indicated that a significant main effect was found for Q:H ($F(1,30) = 17.521$, $p < 0.01$, $\eta^2 = .369$ as well as a significant interaction effect between activation ratio and group ($F(1,30)$, $p = 0.041$, $\eta^2 = .132$) (See Table 5). Eta squared value of 0.132 indicated a medium, approaching strong effect. A simple effect analysis was performed to understand the significant interaction effect using Least Significant Difference (LSD). The results showed that Q:H was significantly higher with forward lunges than backward lunges for those who started backward lunges first ($M_{diff} = 1.88$, SE = 0.43, $p < 0.01$, CI95 [0.99, 2.76]). However, no difference in Q:H was found between forward lunges and backward lunges amongst those who started forward lunges first.

Table 5: Repeated measures ANOVA[a] for quadriceps to hamstrings coactivation ratio between forward and backward lunges by group.

Effect	F	Hypothesis df	Error df	Sig.	Partial Eta Squared
QH_Ratio	17.521[b]	1.000	30.000	< 0.001	0.369
QH_Ratio * Group	4.570[b]	1.000	30.000	0.041	0.132

a. Design: Intercept + Group
b. Exact statistic
Q:H: quadriceps to hamstrings activation ratio.

4.3 Marker-Based Hip and Knee Kinematics

Results from intraclass correlation assessments showed excellent agreement between events defined by kinematic data and events defined by kinetic data. Therefore, lunge events were able to be defined using kinematic-only data.

A comparison of mean hip and knee angles are presented in Tables 6 and 7 for forward and backward lunges, respectively. A side-by-side comparison of max hip angle and total hip range of motion between forward and backward lunges is presented in Table 8 and a side-by-side comparison of max knee angle and total knee angle range of motion between forward and backward lunges is presented in Table 9.

Table 6: Descriptive statistics for hip and knee angles in the forward lunge by group.

Group	Joint	Measurement	Mean	Std. Deviation	Range
Bkwd	Hip	Max	102.175	8.449	26.625
		Min	11.405	5.935	18.091
		ROM	90.770	7.271	24.297
	Knee	Max	116.909	6.192	19.576
		Min	11.954	4.592	14.936
		ROM	104.955	5.399	21.181
Fwd	Hip	Max	95.084	8.335	27.358
		Min	7.577	5.254	15.144
		ROM	87.507	6.338	24.474
	Knee	Max	115.970	7.183	24.388
		Min	9.717	4.320	17.069
		ROM	106.253	8.829	35.015

Fwd: group that performed the forward lunge first, Bkwd: group that performed the backward lunge first, ROM: range of motion (difference between max angle and min angle).

Table 7: Descriptive statistics for hip and knee angles in the backwards lunge by group.

Group	Joint	Measurement	Mean	Std. Deviation	Range
Bkwd	Hip	Max	105.788	10.463	36.979
		Min	12.916	5.529	17.787
		ROM	92.872	9.652	37.660
	Knee	Max	111.565	8.610	27.927
		Min	12.186	5.217	20.034
		ROM	99.378	7.998	23.991
Fwd	Hip	Max	98.569	5.330	19.251
		Min	9.373	5.346	19.129
		ROM	89.197	5.828	24.892
	Knee	Max	109.095	9.924	38.218
		Min	11.380	5.323	18.308
		ROM	97.715	10.620	45.469

Fwd: group that performed the forward lunge first, Bkwd: group that performed the backward lunge first, ROM: range of motion (difference between max angle and min angle).

Table 8: Descriptive statistics for hip angle and range of motion for the forward and backward lunge by group.

Group	Direction	Measurement	Mean	Std. Deviation	Range
Bkwd	Forward	Max	102.175	8.449	26.625
		ROM	90.770	7.271	24.297
	Backward	Max	105.788	10.463	36.979
		ROM	92.872	9.652	37.660
Fwd	Forward	Max	95.084	8.335	27.358
		ROM	87.507	6.338	24.474
	Backward	Max	98.569	5.330	19.251
		ROM	89.197	5.828	24.892

Fwd: group that performed the forward lunge first, Bkwd: group that performed the backward lunge first, ROM: range of motion (difference between max angle and min angle).

Table 9: Descriptive statistics for knee angle and range of motion for the forward and backward lunge by group.

Group	Direction	Measurement	Mean	Std. Deviation	Range
Bkwd	Forward	Max	116.909	6.192	19.576
		ROM	104.955	5.399	21.181
	Backward	Max	111.565	8.610	27.927
		ROM	99.378	7.998	23.991
Fwd	Forward	Max	115.970	7.183	24.388
		ROM	106.253	8.829	35.015
	Backward	Max	109.095	9.924	38.218
		ROM	97.715	10.620	45.469

Fwd: group that performed the forward lunge first, Bkwd: group that performed the backward lunge first, ROM: range of motion (difference between max angle and min angle).

4.3.1 Comparison of Hip Angle

Mean max hip angle was calculated to be 98.51 (SD = 8.99) degrees for the forward lunge and 102.05 (SD = 8.86) degrees for the backwards lunge. Results indicated

that a significant main effect was found for maximal hip angle ($F(1,27) = 17.42$, $p < 0.01$, $n^2 = 0.392$) but no significant interaction effect was found (See Table 10). A partial Eta squared value of 0.392 indicated a strong effect.

Table 10: Repeated measures ANOVA[a] for maximal hip angle for forward and backward lunges by group.

Effect	F	Hypothesis df	Error df	Sig.	Partial Eta Squared
Max Hip Angle	17.424[b]	1.000	27.000	<0.001	0.392
Max Hip Angle * Group	.006[b]	1.000	27.000	0.941	0.000

a. Design: Intercept + Group
b. Exact statistic
df: degrees of freedom

Mean hip range of motion was calculated to be 89.08 (SD = 6.88) degrees for the forward lunge and 90.97 (SD = 7.98) degrees for the backward lunge. Results showed that there was no significant main effect or interaction effect of group or lunge direction on hip range of motion (Table 11).

Table 11: Repeated measures ANOVA[a] for hip range of motion for forward and backward lunges by group.

Effect	F	Hypothesis df	Error df	Sig.	Partial Eta Squared
Hip Range of Motion	3.491[b]	1.000	27.000	0.073	0.114
Hip Range of Motion * Group	.041[b]	1.000	27.000	0.840	0.002

a. Design: Intercept + Group
b. Exact statistic
df: degrees of freedom

4.3.2 Comparison of Knee Angle

Mean max knee angles were calculated to be 116.42 (SD = 6.62) degrees for the forward lunge and 110.29 (SD = 9.23) degrees for the backward lunge. ANOVA results indicated that a significant main effect was found for lunge type on maximal knee angle ($F(1,27) = 22.84$, $p < 0.01$, $n^2 = 0.458$) but no significant interaction effect was found (See Table 12). A partial Eta squared value of 0.458 indicated a strong effect.

Table 12: Repeated measures ANOVA[a] for maximal knee angle for forward and backward lunges by group.

Effect	F	Hypothesis df	Error df	Sig.	Partial Eta Squared
Max Knee Angle	22.835[b]	1.000	27.000	<0.001	0.458
Max Knee Angle * Group	.359[b]	1.000	27.000	0.554	0.013

a. Design: Intercept + Group
b. Exact statistic
df: degrees of freedom

Mean knee range of motion was calculated to be 105 (SD = 7.28) degrees for the forward lunge and 98.52 (SD = 9.32) degrees for the reverse lunge. ANOVA results showed that a significant main effect was found for lunge type on knee angle ($F(1,27) = 23.30$, $p < 0.01$, $n^2 = 0.463$) but no interaction effect was found (Table 13). A partial Eta squared value of 0.463 indicated a strong effect.

Table 13: Repeated measures ANOVA[a] for knee range of motion for forward and backward lunges by group.

Effect	F	Hypothesis df	Error df	Sig.	Partial Eta Squared
Knee Range of Motion	23.297[b]	1.000	27.000	<0.001	0.463
Knee Range of Motion * Group	1.026[b]	1.000	27.000	0.320	0.037

a. Design: Intercept + Group
b. Exact statistic
df: degrees of freedom

4.4 Validation of Markerless Calculation of Hip and Knee Kinematics

Results for joint kinematics as measured by the markerless system are shown in Table 14. Overall, ICCs for joint kinematics ranged from moderate to good agreement. Forward lunging consistently showed moderate to good agreement across joints, whereas backward lunging showed good to excellent agreement.

Table 14: Two-way mixed effects, interclass correlation coefficient for absolute agreement between maximum joint angles at the bottom of the lunge and total range of motion calculated from the markerless tracking.

Direction	Joint	Variable	Mean ± SE	Intraclass Correlation	[CI 95%]
Forward	Hip	Max Angle	92.513 ± 1.63	.771**	[.533, .896]
		Total ROM	84.334 ±1.542	.762**	[.518, .892]
	Knee	Max Angle	109.081 ± 1.020	.862**	[.702, .939]
		Total ROM	101.321 ± 1.068	.508*	[.130, .757]
Backward	Hip	Max Angle	95.750 ± 1.476	.739**	[.477, .880]
		Total ROM	88.097 ± 1.506	.725**	[.454, .873]
	Knee	Max Angle	102.597 ± 1.827	.957***	[.900, .981]
		Total ROM	93.559 ± 2.018	.914***	[.807, .962]
Valid N (listwise)	23				

* Indicates moderate agreement, ** indicates good agreement, and *** indicates excellent agreement between systems. SE: Standard Error, CI95%: 95% Confidence Interval

Based on these results, mixed-design ANOVA tests were run to further investigate differences between systems. Due to the lack of interaction effects found between lunges for the marker-based system, only interaction effects for motion capture system were investigated.

4.4.1 Maximal Hip Angle

Mean max hip angles were calculated to be 98.80 (SD = 9.23) degrees for the forward lunge and 102.18 (SD = 9.07) degrees for the backward lunge using the marker-based system and 86.23 (SD = 8.00) degrees for the forward lunge and 89.31 (SD = 7.38) degrees for the backward lunge using the markerless system. ANOVA results indicated that a significant main effect was found for lunge type on maximal hip angle ($F(1,22) = 11.889$, $p = 0.002$, $n^2 = 0.351$) as well as system (($F(1,22) = 116.215$, $p < 0.001$, $n^2 = 0.841$) but no interaction effect was seen (see Table 15). A partial Eta squared value of 0.351 for lunge type indicated a strong effect, whereas eta squared value of 0.841 indicated a strong effect for system.

Table 15: Repeated measures ANOVA[a] for maximal hip angle for forward and backward lunge by system.

Effect	F	Hypothesis df	Error df	Sig.	Partial Eta Squared
Max Hip Angle for Forward vs Backward Lunge	11.889[b]	1.000	22.000	0.002	0.351
Marker-based vs Markerless System	116.215[b]	1.000	22.000	<.001	0.841

a. Design: Intercept
Within Subjects Design: fwd_bkwd + system + fwd_bkwd * system

b. Exact statistic

4.4.2 Total Hip Range of Motion

Mean hip range of motion was calculated to be 89.36 (SD = 7.56) degrees for the forward lunge and 90.41 (SD = 8.71) degrees for the backward lunge using the marker-based system and 79.30 (SD = 7.80) degrees for the forward lunge and 85.79 (SD = 7.59) degrees for the backward lunge using the markerless system. ANOVA results indicated that a significant main effect was found for lunge type on maximal hip angle ($F(1,22) = 15.561$, $p < 0.001$, $n^2 = 0.414$) as well as system (($F(1,22) = 46.727$, $p < 0.001$, $n^2 = 0.680$) and a significant interaction effect between lunge direction and system ($F(1,30) = 29.509$, $p < 0.001$, $\eta^2 = .573$) (see Table 16). A partial Eta squared value of 0.414 for lunge type indicated a strong effect, a partial eta squared value of 0.680 indicated a strong effect for system, and a partial eta squared value of 0.573 indicated a strong effect for the interaction between lunge direction and system. A simple effect was performed to understand significant interaction effect using LSD. The results showed that the markerless system measured a significant difference in hip range of motion between lunges ($M_{diff} = -6.490$, SE = 0.973, $p < 0.001$, CI95 [-8.507, -4.472]); however, no difference in hip range of motion between lunges was found by the marker-based system.

Table 16: Repeated measures ANOVA[a] for total hip range of motion for forward and backward lunges by system.

Effect	F	Hypothesis df	Error df	Sig.	Partial Eta Squared
Total Hip Range of Motion for Forward vs Backward Lunge	15.561[b]	1.000	22.000	<.001	0.414
Marker-based vs Markerless System	46.727[b]	1.000	22.000	<.001	0.680
Interaction Effect	29.509[b]	1.000	22.000	<.001	0.573

a. Design: Intercept
b. Exact statistic

4.4.3 Maximal Knee Angle

Mean max knee angles were calculated to be 116.50 (SD = 6.37) degrees for the forward lunge and 109.94 (SD = 10.17) degrees for the backward lunge using the marker-based system and 101.66 (SD = 5.06) degrees for the forward lunge and 95.26 (SD = 9.31) degrees for the backwards lunge using the markerless system. ANOVA results indicated that a significant main effect was found for lunge type on maximal knee angle ($F(1,22) = 20.518$, $p < 0.001$, $n^2 = 0.483$) as well as system (($F(1,22) = 611.108$, $p < 0.001$, $n^2 = 0.965$) but no interaction effect was found (see Table 17). A parital Eta squared value of 0.483 for lunge type indicated a strong effect, whereas eta squared value of 0.965 indicated a strong effect for system.

Table 17: Repeated measures ANOVA[a] for maximal knee angle for forward and backward lunges by system.

Effect	F	Hypothesis	Error df	Sig.	Partial Eta
Max Knee Angle for Forward vs Backward Lunge	20.518[b]	1.000	22.000	<.001	0.483
Marker-based vs Markerless System	611.108[b]	1.000	22.000	<.001	0.965

a. Design: Intercept
Within Subjects Design: fwd_bkwd + system + fwd_bkwd * system

b. Exact statistic

4.4.4 Total Knee Range of Motion

Mean knee range of motion was calculated to be 105.42 (SD = 6.71) degrees for the forward lunge and 97.49 (SD = 9.68) degrees for the reverse lunge using the marker-based system and 97.22 (SD = 5.45) degrees for the forward lunge and 89.62 (SD = 10.48) degrees for the backward lunge using the markerless system. ANOVA results indicated that a significant main effect was found for lunge type on total knee range of motion ($F(1,22) = 20.246$, $p < 0.001$, $n^2 = 0.479$) as well as system (($F(1,22) = 62.957$, $p < 0.001$, $n^2 = 0.741$ but no significant interaction effect was seen (see Table 22). A partial Eta squared value of 0.479 for lunge type indicated a strong effect, whereas eta squared value of 0.741 indicated a strong effect for system.

Table 18: Repeated measures ANOVA[a] for total knee range of motion for forward and backward lunges by system.

Effect	F	Hypothesis df	Error df	Sig.	Partial Eta Squared
Max Hip Angle for Forward vs Backward Lunge	20.246[b]	1.000	22.000	<.001	0.479
Marker-based vs Markerless System	62.597[b]	1.000	22.000	<.001	0.741

a. Design: Intercept
Within Subjects Design: fwd_bkwd + system + fwd_bkwd * system

b. Exact statistic

CHAPTER 5: DISCUSSION

5.1 Electromyography

The first aim of this dissertation was to examine the difference between muscle activation ratios of the quadriceps and hamstrings muscles during the forward and backward lunge exercise. The results from this experiment showed that the forward lunge produced a Q:H of 6.83, whereas the backward lunge produced a Q:H of 5.63, which was shown to be a statistically significant difference (See Tables 5 and 6). This means that the quadriceps activity was higher when performing the forward lunge than the reverse lunge. Additionally, an interaction effect was seen when activation ratios were compared by groups. The group that performed the backward lunge first had a 1.88 mV difference between the forward and backward lunge.

Further analysis suggests that the difference in activation ratios between lunges can be explained by the difference in quadriceps muscle activity. The results of the ANOVA from Table 3 showed that the forward lunge produced significantly greater muscle activity than the backward lunge, whereas the analysis of hamstrings activity in Table 4 showed no difference between lunge type.

These results appear to be consistent with previous research that examined quadriceps activity in the lunge. Park and colleagues demonstrated that electrical activity from the vastus medialis oblique muscle was significantly higher when performing the

forward lunge, compared to the backwards lunge (Park et al., 2021). Additionally, Goulette et al. reported that the forward lunge resulted in higher patellofemoral joint loading, compared to the backward lunge (Goulette et al., 2021). They calculated the loading from the forces generated by the rectus femoris, vastus lateralis, and vastus intermedius using the Human Body Model (Motek Medical, Amsterdam, The Netherlands) and musculoskeletal modeling processes published by van den Bogert and colleagues (Van Den Bogert, Geijtenbeek, Even-Zohar, Steenbrink, & Hardin, 2013). This study examined the quadriceps activity as a whole and saw that performing the forward lunge resulted in approximately 24% higher EMG values when compared to the backward lunge.

A study conducted by Krause et al., showed no difference in rectus femoris activity between the forward lunge and the lunge with the rear foot suspended, but nearly twice as much activity in the hamstrings and gluteals during the rear foot suspended lunge (Krause et al., 2018). In our study, we found that there was no significant difference in hamstring activation between lunge variations. This could be due to differences in the nature of the movement. In the rear foot suspended lunge, the back leg would be unstable and, therefore unable to contribute significantly to force production. Additionally, the subject's center of mass moves posterior to the lead leg's base of support, which means that the hamstrings would be recruited to a greater degree to "pull" the subject up to the standing position. This logic appears to be consistent with research recently conducted by Demir and Harput, who showed that hamstrings muscle activity was greatly increased

when performing the return phase of a forward lunge on a slideboard (Demir & Harput, 2023).

Since the hamstrings activity was not significantly different between lunge variations, the difference in quadriceps activity is likely the reason for the statistically different Q:H. Our study showed that the Q:H for the forward lunge was 6.83 and 5.63 for the backward lunge. While not as large as the value found by Begalle and colleagues, our results are still consistent in that the activation ratio was significantly elevated in the forward lunge (Begalle et al., 2012).

Current ACL rehabilitation protocols focus on the individual patient's progress, rather than establishing strict procedures and timelines (Van Grinsven, Van Cingel, Holla, & Van Loon, 2010). Goals are set based on a combination of range of motion, strength, balance gains surrounding the affected limb, and the ability to achieve improvements in pain-free movement. The recommendation for improving strength is to increase quadriceps activation early in the rehabilitation process to regain patellofemoral joint control and then increase difficulty as tolerated by the patient (Cavanaugh & Powers, 2017). While previous research has identified the risk of a large imbalance in quadriceps and hamstrings coactivation on ACL re-injury, no research has identified an optimal timeframe for the integration of different therapeutic exercises based on their coactivation ratios (Begalle et al., 2012). Further research on therapeutic exercise prescription for ACL rehabilitation should focus on identifying coactivation ratios and making

recommendations on when the exercises should be integrated to both optimize leg strengthening while concurrently minimizing the risk of ACL reinjury.

The other significant finding of this study was the interaction effect seen between lunge variations in the group that performed the backwards lunge first. We can only identify two theories that could explain these findings. First, all groups performed their familiarization lunges by going forward first and then backward. It is possible that there could have been residual fatigue from the familiarization set that could have reduced the quadriceps activity. The group that performed the forward lunge first did not have any difference in quadriceps activation, possibly due to residual fatigue. By performing the backward lunge first, the reduced workload on the quadriceps could have allowed for more recovery, which would have resulted in greater electrical activity when the forward lunge was performed. Second, it was noticed that several of the subjects were not as familiar with the movement of the backward lunge as they were with the forward lunge. It is possible that by performing the forward lunge first, it allowed extra practice repetitions and, specifically, familiarization of the bottom position of the lunge. The unfamiliarity of the backwards lunge and consequent instability when performing the movement could have had an interaction effect that we did not adjust for, which could have impacted the amount of muscle activity.

There are several limitations to this study. First, the inherent limitations with using surface electromyography should be noted. When using surface electromyography,

it is always likely that noise from adjacent muscles will be detected and recorded as activity of the target muscle, commonly referred to as “crosstalk.”

Second, slight variations in site preparation and sensor placement could contribute to any presence of a Type II error. Although only one technician was responsible for the site preparation and sensor placement for all subjects, variations from subject to subject could contribute to human error.

Third, our equipment was unable to time normalize our data. As a result, we were only able to identify the highest EMG amplitude without consideration of where exactly the peak activity occurred in the lunge. This means that we could not identify the joint angles that correspond with the highest activation of muscle activity, nor could we identify when peak activity of each individual muscle occurred relative to the other muscles. By time normalizing the data and being able to identify corresponding joint angles, position-specific data can be reported and determined to see if differences exist in hip and knee joint angle can explain the variation seen in EMG data.

Finally, subjects’ unfamiliarity of the backwards lunge could have influenced muscle activity. Subsequent research should limit the subject pool to only include those who regularly incorporate backward lunges into their exercise routines.

5.2 Marker-Based Hip and Knee Kinematics

The purpose of this study was to examine the differences in hip and knee joint angle between the forward and backward lunge. Since no interaction effect was observed between lunge groups, all data were interpreted as differences between forward and backward lunges.

The results of this investigation showed that the forward lunge resulted in a hip angle of 98.51 degrees and a knee angle of 116.42 degrees, whereas the backward lunge resulted in a hip angle of 102.05 degrees and a knee angle of 110.29 degrees. These results seem consistent with the findings from the previous aim, which showed that the forward lunge resulted in greater quadriceps activity. Since the knee angle was approximately 6 degrees greater in the forward lunge, this could have resulted in the increased quadriceps activity. Further analysis did not show a significant correlation between quadriceps activity and maximum knee joint angle. However, after visualization of the data, it appears that the presence of outliers may affect the correlation between joint angle and quadriceps activity. Further research should increase the sample size to elucidate this theory.

It should also be noted that total hip and knee range of motion for both lunges were analyzed. Results indicated that total hip range of motion was not statistically different between lunge types, but knee range of motion was approximately 7 degrees higher in the forward lunge than the backward lunge. This supports the theory that the

increased knee angle in the forward lunge resulted in higher EMG activity in the quadriceps.

To our knowledge, this is the first study to examine the difference in hip and knee joint angles between two lunge variations. Previous research has established that varying exercise form or load placement can impact joint angles of the lower extremities and consequently, muscle activity (Escamilla et al., 2001; Hemmerich, Brown, Smith, Marthandam, & Wyss, 2006; Slater & Hart, 2017; Swinton, Lloyd, Keogh, Agouris, & Stewart, 2012). Additionally, previous work has specifically examined the forward lunge and showed that variations in step length and torso angle can influence joint angle and muscle activity levels (Escamilla et al., 2022; Farrokhi et al., 2008). In our study, we found differences in both hip and knee max angles, which occurred at the bottom of the lunge. We also found a corresponding difference in muscle activity with the lunge variations.

There are a few limitations to note in this study. First, the analysis showed that there was a difference in hip max angle but not total hip range of motion. This means that for the forward lunge, although the max hip angle was greater than the backward lunge, the total degrees traveled was not, meaning that the subjects started in a more flexed position at the beginning of the movement. This was not seen at the knee joint, which showed a difference in max knee angle and total range of motion. Further research should

seek to address this potential error and ensure that all subjects begin in a fully hip and knee extended position at the beginning of the trial.

Second, spinal/torso angle was not included in the analysis, therefore it is unknown whether differences in hip angle could be attributed to the angle of the spine. Previous research has shown how altering the torso angle when performing a lunge will change the degree of hip flexion and concurrent muscle activity (Farrokhi et al., 2008). Follow up research should track movement of the torso and attempt to keep torso angle consistent across lunge types to see if the direction of the lunge influences joint angle.

Finally, the filtering rate for the motion capture data was set at 1Hz, due to the low velocity nature of the movement. This was done in an attempt to eliminate any spikes or artifacts. It is possible that the true joint max values were filtered out if they occurred within the 1 Hz window. Research should continue to validate the filtering of low velocity data at lower frequencies to determine if this is a viable means of assessing discreet values like maximal joint angles.

5.3 Markerless Hip and Knee Kinematics

The purpose of this study was to examine hip and knee kinematics of the forward and backward lunge using a markerless motion capture system and compare it to the gold standard marker-based system.

The results of this research aim showed moderate to good agreement between systems when subjects performed the forward lunge. Similarly, the backward lunge showed good to excellent agreement across all measured variables. Further investigation of differences showed that significant differences in calculated hip and knee joint angles and total range of motion at the hip and knee were found between systems across lunge types.

At the hip, the difference in total range of motion at the hip between the forward and backward lunge was calculated to be -6.49 degrees using the markerless system, indicating that the hip range of motion of the backwards lunge was determined to be larger. Conversely, the marker-based system did not find a significant difference between ranges of motion.

At the knee, the difference in max joint angle at the bottom of the lunge between the forward and the backward lunges as well as total range of motion was found to be significantly different. The measured differences between systems were also found to vary significantly; however, no significant effect was found between system and lunge type.

In this study, we found that a significant difference in was found between systems. One possible explanation could be due to the lack of training of the current model on the lunge. In order for artificial intelligence-based systems to be effective in pose estimation, they require sample data to build deep convolutional neural networks (CNN) and effectively identify body segments (Chen et al., 2016; Mathis et al., 2018). To date, most research with KinaTrax markerless motion capture has been conducted on gait (Ripic, Nienhuis, et al., 2023; Ripic, Signorile, et al., 2023; Ripic et al., 2022) or baseball pitching and batting (Yeadon & Pain, 2023); however, no research has been conducted using the lunge exercise. Previous research that examined the accuracy of this system found reduced reliability in properly tracking the pelvis during gait (Ripic, Nienhuis, et al., 2023). This could be due to the difficulty in properly identifying joint centers in larger body segments, which could explain reduced agreement between systems.

There were several limitations present in this study. First, the motion capture systems were not integrated and, therefore the data could not be time normalized. Additionally, KinaTrax was not connected to force plates, meaning kinetic data, which is traditionally used to identify events, was not able to be used. This means that kinematic events had to be used to identify the beginning, bottom, and end of the lunge trials, which could introduce a source of error if these events were not properly defined. Another limitation to this study was that subjects only came in on a single visit to perform their lunges. Having repeated sessions may have provided more data points and improved agreement between systems. Fourth, similar to the previous aim, frame filtering was set at 1Hz to remove any potential impulses or artifacts. It is possible that this could have

affected the validity of the markerless results. Finally, this study was conducted using only healthy young adults, thereby reducing the external validity of this research.

5.4 Conclusion

In this dissertation, the differences in muscle activation and joint angles between forward and backward lunges were investigated. Additionally, a commercially available 3-dimensional markerless motion capture system was assessed for its ability to accurately measure joint angles in these lunge variations, compared to a gold standard motion capture system. For muscular activity, it was found that hamstrings activity was not different between lunges, but quadriceps activity was significantly higher in the forward lunge than in the backward lunge, leading to a significantly higher quadriceps to hamstrings coactivation ratio.

For hip and knee joint angles, it was found that maximal knee joint angles were larger in the forward lunge, compared to the backward lunge. It was also found that total hip range of motion was not different between lunge types.

For the validation of the markerless motion capture system, agreement ranged from moderate to excellent, with the forward lunge showing better agreement than the backward lunge. Additionally, a significantly higher difference in hip joint angle was

found between the forward and backward lunge as compared to the marker-based system, which did not find as large of a difference.

Additional research in this field should continue to evaluate muscular activation differences between these exercises and work to identify optimal coactivation ratios for rehabilitation protocols. Additionally, improvements should be made to improve the accuracy of the markerless system, particularly in its ability to accurately measure backwards movement.

REFERENCES

Adrian, E. D., & Bronk, D. W. (1929). The discharge of impulses in motor nerve fibres: Part II. The frequency of discharge in reflex and voluntary contractions. *Journal of Physiology, 67*(2), i3.

Ahmad, C. S., Clark, A. M., Heilmann, N., Schoeb, J. S., Gardner, T. R., & Levine, W. N. (2006). Effect of gender and maturity on quadriceps-to-hamstring strength ratio and anterior cruciate ligament laxity. *The American Journal of Sports Medicine, 34*(3), 370-374.

Alentorn-Geli, E., Myer, G. D., Silvers, H. J., Samitier, G., Romero, D., Lázaro-Haro, C., & Cugat, R. (2009). Prevention of non-contact anterior cruciate ligament injuries in soccer players. Part 1: Mechanisms of injury and underlying risk factors. *Knee Surgery, Sports Traumatology, Arthroscopy, 17*, 705-729.

Andriacchi, T. P., & Alexander, E. J. (2000). Studies of human locomotion: past, present and future. *Journal of Biomechanics, 33*(10), 1217-1224.

Armitano-Lago, C., Willoughby, D., & Kiefer, A. W. (2021). A SWOT Analysis of Portable and Low-Cost Markerless Motion Capture Systems to Assess Lower-Limb Musculoskeletal Kinematics in Sport. *Frontiers in Sports and Active Living, 3*, 809898. doi:10.3389/fspor.2021.809898

Banta, J. (1999). Gait analysis: past, present, and future. *Developmental Medicine and Child Neurology, 41*(6), 363. doi:10.1017/s001216229900081x

Barringer, J. M. (2014). *The art and archaeology of ancient Greece*. Cambridge: Cambridge University Press.

Barton, C. J., Lack, S., Malliaras, P., & Morrissey, D. (2013). Gluteal muscle activity and patellofemoral pain syndrome: a systematic review. *British Journal of Sports Medicine, 47*(4), 207-214.

Baumgart, C., Schubert, M., Hoppe, M. W., Gokeler, A., & Freiwald, J. (2017). Do ground reaction forces during unilateral and bilateral movements exhibit compensation strategies following ACL reconstruction? *Knee Surgery, Sports Traumatology, Arthroscopy, 25*, 1385-1394.

Begalle, R. L., DiStefano, L. J., Blackburn, T., & Padua, D. A. (2012). Quadriceps and hamstrings coactivation during common therapeutic exercises. *Journal of Athletic Training, 47*(4), 396-405.

Behm, D. G., Leonard, A. M., Young, W. B., Bonsey, W. A. C., & MacKinnon, S. N. (2005). Trunk muscle electromyographic activity with unstable and unilateral exercises. *The Journal of Strength & Conditioning Research, 19*(1), 193-201.

Beynnon, B. D., Fleming, B. C., Johnson, R. J., Nichols, C. E., Renström, P. A., & Pope, M. H. (1995). Anterior cruciate ligament strain behavior during rehabilitation exercises in vivo. *The American Journal of Sports Medicine, 23*(1), 24-34.

Bini, R. R., Diefenthaeler, F., & Mota, C. B. (2010). Fatigue effects on the coordinative pattern during cycling: Kinetics and kinematics evaluation. *Journal of Electromyography and Kinesiology, 20*(1), 102-107.

Boden, B. P., Dean, G. S., Feagin, J. A., & Garrett, W. E. (2000). Mechanisms of anterior cruciate ligament injury. In (Vol. 23, pp. 573-578): SLACK Incorporated Thorofare, NJ.

Bollen, S. (2000). Epidemiology of knee injuries: diagnosis and triage. *British Journal of Sports Medicine, 34*(3), 227-228.

Boudreau, S. N., Dwyer, M. K., Mattacola, C. G., Lattermann, C., Uhl, T. L., & McKeon, J. M. (2009). Hip-muscle activation during the lunge, single-leg squat, and step-up-and-over exercises. *Journal of Sport Rehabilitation, 18*(1), 91-103.

Brazier, M. A. (2012). The emergence of electrophysiology as an aid to neurology. *Aminoff's Electrodiagnosis in Clinical Neurology E-Book: Expert Consult-Online and Print*, 1.

Bregler, C. (2007). Motion capture technology for entertainment [in the spotlight]. *IEEE Signal Processing Magazine, 24*(6), 160-158.

Campanini, I., Disselhorst-Klug, C., Rymer, W. Z., & Merletti, R. (2020). Surface EMG in Clinical Assessment and Neurorehabilitation: Barriers Limiting Its Use. *Frontiers in Neurology, 11*. doi:10.3389/fneur.2020.00934

Cavanaugh, J. T., & Powers, M. (2017). ACL rehabilitation progression: where are we now? *Current Reviews in Musculoskeletal Medicine, 10*, 289-296.

Chandler, J., McMillan, J., Kibler, B., & Richards, D. (2002). Safety of the squat exercise. *ACSM Current Comments*. Retrieved from: https://bpb-us-e2.wpmucdn.com/websites.umass.edu/dist/a/2523/files/2009/07/safetysquatexercise.pdf

Chek, P. (2000). Movement that matters. *San Diego, CA: CHEK Institute, 54. ISBN: 1583870024*.

Chen, W., Wang, H., Li, Y., Su, H., Wang, Z., Tu, C., . . . Chen, B. (2016). *Synthesizing training images for boosting human 3d pose estimation.* Paper presented at the 2016 Fourth International Conference on 3D Vision (3DV).

Clayton, M., & Philo, R. (2010). *Leonardo da Vinci: the mechanics of man*: Getty Publications.

Cohen, J., Cohen, P., West, S. G., & Aiken, L. S. (2013). *Applied multiple regression/correlation analysis for the behavioral sciences*: Routledge.

Cook, G. (2003). *Athletic body in balance*: Human kinetics, 232. ISBN: 9781492583776.

Cram, J. R. (2003). The History of Surface Electromyography. *Applied Psychophysiology and Biofeedback, 28*(2), 81-91. doi:10.1023/A:1023802407132

Criswell, E. (2010). *Cram's introduction to surface electromyography*: Jones & Bartlett Publishers.

Da Vinci, L. (2013). *Leonardo on the human body*: Courier Corporation.

Delp, S. L., Hess, W. E., Hungerford, D. S., & Jones, L. C. (1999). Variation of rotation moment arms with hip flexion. *Journal of Biomechanics, 32*(5), 493-501.

Demir, C., & Harput, G. (2023). Weight-Bearing Exercises on Slideboard Increase Quadriceps and Hamstring Activation Levels and Improve Hip-and Knee-Flexion Angles in Physically Active Individuals. *Journal of Sport Rehabilitation, 1*(aop), 1-13.

Distefano, L. J., Blackburn, J. T., Marshall, S. W., & Padua, D. A. (2009). Gluteal Muscle Activation During Common Therapeutic Exercises. *Journal of Orthopaedic & Sports Physical Therapy, 39*(7), 532-540. doi:10.2519/jospt.2009.2796

Dobos, T. J., Bench, R. W. G., McKinnon, C. D., Brady, A., Boddy, K. J., Holmes, M. W. R., & Sonne, M. W. L. (2022). Validation of pitchAITM markerless motion capture using marker-based 3D motion capture. *Sports Biomechanics*, 1-21. doi:10.1080/14763141.2022.2137425

Draganich, L., Jaeger, R., & Kralj, A. (1989). Coactivation of the hamstrings and quadriceps during extension of the knee. *JBJS, 71*(7), 1075-1081.

Duchenne, G.-B. (1855). *De l'électrisation localisée et de son application à la physiologie, à la pathologie et à la thérapeutique*: Baillière.

Eichelberger, P., Ferraro, M., Minder, U., Denton, T., Blasimann, A., Krause, F., & Baur, H. (2016). Analysis of accuracy in optical motion capture – A protocol for

laboratory setup evaluation. *Journal of Biomechanics, 49*(10), 2085-2088. doi:https://doi.org/10.1016/j.jbiomech.2016.05.007

Ekstrom, R. A., Donatelli, R. A., & Carp, K. C. (2007). Electromyographic Analysis of Core Trunk, Hip, and Thigh Muscles During 9 Rehabilitation Exercises. *Journal of Orthopaedic & Sports Physical Therapy, 37*(12), 754-762. doi:10.2519/jospt.2007.2471

Escamilla, R. F., Fleisig, G. S., Zheng, N., LANDER, J. E., Barrentine, S. W., ANDREWS, J. R., . . . MOORMAN III, C. T. (2001). Effects of technique variations on knee biomechanics during the squat and leg press. *Medicine and Science in Sports and Exercise, 33*(9), 1552-1566.

Escamilla, R. F., Zheng, N., MacLeod, T. D., Imamura, R., Wang, S., Wilk, K. E., . . . Fleisig, G. S. (2022). Patellofemoral Joint Loading in Forward Lunge With Step Length and Height Variations. *Journal of Applied Biomechanics, 38*(4), 210-220.

Farrokhi, S., Pollard, C. D., Souza, R. B., Chen, Y.-J., Reischl, S., & Powers, C. M. (2008). Trunk position influences the kinematics, kinetics, and muscle activity of the lead lower extremity during the forward lunge exercise. *Journal of Orthopaedic & Sports Physical Therapy, 38*(7), 403-409.

Faul, F., Erdfelder, E., Buchner, A., & Lang, A.-G. (2009). Statistical power analyses using G* Power 3.1: Tests for correlation and regression analyses. *Behavior Research Methods, 41*(4), 1149-1160.

Fountaine, C. J. (2018). Unilateral and bilateral exercise movements: Considerations for Program Design. *ACSM's Health & Fitness Journal, 22*(3), 11-16.

Gadomski, S. J., Ratamess, N. A., & Cutrufello, P. T. (2018). Range of Motion Adaptations in Powerlifters. *The Journal of Strength & Conditioning Research, 32*(11), 3020-3028. doi:10.1519/jsc.0000000000002824

Goulette, D., Griffith, P., Schiller, M., Rutherford, D., & Kernozek, T. W. (2021). Patellofemoral joint loading during the forward and backward lunge. *Physical Therapy in Sport, 47*, 178-184.

Gribble, P. A., Brigle, J., Pietrosimone, B. G., Pfile, K. R., & Webster, K. A. (2013). Intrarater Reliability of the Functional Movement Screen. *The Journal of Strength & Conditioning Research, 27*(4), 978-981. doi:10.1519/JSC.0b013e31825c32a8

Hale, T. (2008). History of developments in sport and exercise physiology: AV Hill, maximal oxygen uptake, and oxygen debt. *Journal of Sports Sciences, 26*(4), 365-400.

Hefzy, M., Al Khazim, M., & Harrison, L. (1997). Co-activation of the hamstrings and quadriceps during the lunge exercise. *Biomedical Sciences Instrumentation, 33*, 360-365.

Hemmerich, A., Brown, H., Smith, S., Marthandam, S., & Wyss, U. (2006). Hip, knee, and ankle kinematics of high range of motion activities of daily living. *Journal of Orthopaedic Research, 24*(4), 770-781.

Hooker, J. T. (1980). The Ancient Spartans, 254. ISBN: 0460043528.

Hug, F. (2011). Can muscle coordination be precisely studied by surface electromyography? *Journal of Electromyography and Kinesiology, 21*(1), 1-12. doi:https://doi.org/10.1016/j.jelekin.2010.08.009

Inman, V. T., Ralston, H. J., & Todd, F. (1981). *Human Walking* (J. C. Leiberman Ed.): Williams & Wilkins.

Kanko, R. M., Laende, E. K., Strutzenberger, G., Brown, M., Selbie, W. S., DePaul, V., . . . Deluzio, K. J. (2021). Assessment of spatiotemporal gait parameters using a deep learning algorithm-based markerless motion capture system. *Journal of Biomechanics, 122*, 110414. doi:10.1016/j.jbiomech.2021.110414

Kazamel, M., & Warren, P. P. (2017). History of electromyography and nerve conduction studies: A tribute to the founding fathers. *Journal of Clinical Neuroscience, 43*, 54-60.

Keogh, J. (1999). Lower-Body Resistance Training: Increasing Functional Performance with Lunges. *Strength & Conditioning Journal, 21*(1), 67. Retrieved from https://journals.lww.com/nsca-scj/Fulltext/1999/02000/Lower_Body_Resistance_Training__Increasing.17.aspx

Kleissen, R., Buurke, J., Harlaar, J., & Zilvold, G. (1998). Electromyography in the biomechanical analysis of human movement and its clinical application. *Gait & Posture, 8*(2), 143-158.

Koo, T. K., & Li, M. Y. (2016). A Guideline of Selecting and Reporting Intraclass Correlation Coefficients for Reliability Research. *Journal of Chiropractic Medicine, 15*(2), 155-163. doi:https://doi.org/10.1016/j.jcm.2016.02.012

Kotsifaki, A., Whiteley, R., & Hansen, C. (2018). Dual Kinect v2 system can capture lower limb kinematics reasonably well in a clinical setting: concurrent validity of a dual camera markerless motion capture system in professional football players. *BMJ Open Sport & Exercise Medicine, 4*(1). doi:ARTN e00044110.1136/bmjsem-2018-000441

Kraus, K., Schütz, E., Taylor, W. R., & Doyscher, R. (2014). Efficacy of the functional movement screen: a review. *The Journal of Strength & Conditioning Research, 28*(12), 3571-3584.

Krause, D. A., Elliott, J. J., Fraboni, D. F., McWilliams, T. J., Rebhan, R. L., & Hollman, J. H. (2018). Electromyography of the hip and thigh muscles during two variations of the lunge exercise: a cross-sectional study. *International Journal of Sports Physical Therapy, 13*(2), 137.

Krautmann, A. C., & Ciecka, J. (2009). The Postseason Value of an Elite Player to a Contending Team. *Journal of Sports Economics, 10*(2), 168-179. doi:10.1177/1527002508321457

Kritz, M., Cronin, J., & Hume, P. (2009). Using the body weight forward lunge to screen an athlete's lunge pattern. *Strength & Conditioning Journal, 31*(6), 15-24.

Marchetti, P. H., Guiselini, M. A., Silva, J. J. d., Tucker, R., Behm, D. G., & Brown, L. E. (2018). Balance and Lower Limb Muscle Activation Between in-Line and Traditional Lunge Exercises. *Journal of Human Kinetics, 62*(1), 15-22. doi:doi:10.1515/hukin-2017-0174

Mathis, A., Mamidanna, P., Cury, K. M., Abe, T., Murthy, V. N., Mathis, M. W., & Bethge, M. (2018). DeepLabCut: markerless pose estimation of user-defined body parts with deep learning. *Nature Neuroscience, 21*(9), 1281-1289.

McClellan, T., & Bugg, B. S. (1999). Lunge Variations to Enhance Specificity in Tennis. *Strength & Conditioning Journal, 21*(6), 18. Retrieved from https://journals.lww.com/nsca-scj/Fulltext/1999/12000/Lunge_Variations_to_Enhance_Specificity_in_Tennis.5.aspx

McComas, A. (2011). *Galvani's spark: The story of the nerve impulse*: Oxford University Press.

Minick, K. I., Kiesel, K. B., Burton, L., Taylor, A., Plisky, P., & Butler, R. J. (2010). Interrater Reliability of the Functional Movement Screen. *The Journal of Strength & Conditioning Research, 24*(2), 479-486. doi:10.1519/JSC.0b013e3181c09c04

Mitchell, H. (1952). Sparta London. In: Cambridge University Press.

Montalvo, A. M., Schneider, D. K., Yut, L., Webster, K. E., Beynnon, B., Kocher, M. S., & Myer, G. D. (2019). “What’s my risk of sustaining an ACL injury while playing sports?” A systematic review with meta-analysis. *British Journal of Sports Medicine, 53*(16), 1003-1012.

Moore, D., Semciw, A. I., & Pizzari, T. (2020). A systematic review and meta-analysis of common therapeutic exercises that generate highest muscle activity in the gluteus medius and gluteus minimus segments. *International Journal of Sports Physical Therapy, 15*(6), 856.

Moran, R. W., Schneiders, A. G., Mason, J., & Sullivan, S. J. (2017). Do Functional Movement Screen (FMS) composite scores predict subsequent injury? A systematic review with meta-analysis. *British Journal of Sports Medicine, 51*(23), 1661-1669.

Mundermann, L., Corazza, S., & Andriacchi, T. P. (2006). The evolution of methods for the capture of human movement leading to markerless motion capture for biomechanical applications. *Journal of NeuroEngineering and Rehabilitation, 3*, 6. doi:10.1186/1743-0003-3-6

Nadzalan, A. M., Shafiee, M. S., Mohamad, M. H., Tan, K., Rahman, R. I. A., Mohamad, N. I., & Pratama, R. S. (2021). *The Effects of Loadings during Forward Lunge on Force Output in Dominant and Non-Dominant Leg.* Paper presented at the Journal of Physics: Conference Series.

Nebel, A. R. (2022). *Predictors of Peak Elbow Valgus Torque in Collegiate Baseball Pitchers.* (M.S.). University of Arkansas, United States -- Arkansas. Retrieved from https://www.proquest.com/docview/2701114279?accountid=14585&parentSessionId=o%2BJPtK44SBSIwek%2F72XM2LQUgZo1La2eAieU2iTT8bM%3D&sourcetype=Dissertations%20&%20Theses ProQuest Dissertations & Theses Global database. (29170561)

Padua, D. A., Marshall, S. W., Boling, M. C., Thigpen, C. A., Garrett Jr, W. E., & Beutler, A. I. (2009). The Landing Error Scoring System (LESS) is a valid and reliable clinical assessment tool of jump-landing biomechanics: the JUMP-ACL study. *The American Journal of Sports Medicine, 37*(10), 1996-2002.

Park, S., Huang, T., Song, J., & Lee, M. (2021). Comparative Study of the Biomechanical Factors in Range of Motion, Muscle Activity, and Vertical Ground Reaction Force between a Forward Lunge and Backward Lunge. *Physical Therapy Rehabilitation Science, 10*(2), 98-105.

Pincivero, D. M., Aldworth, C., Dickerson, T., Petry, C., & Shultz, T. (2000). Quadriceps-hamstring EMG activity during functional, closed kinetic chain exercise to fatigue. *European Journal of Applied Physiology, 81*, 504-509.

Remak, R. (1858). *Galvanotherapie der nerven-und muskelkrankheiten*: Hirschwald.

Riemann, B., Congleton, A., Ward, R., & Davies, G. J. (2013). Biomechanical comparison of forward and lateral lunges at varying step lengths. *Journal of Sports Medicine and Physical Fitness, 53*(2), 130-138.

Ripic, Z., Nienhuis, M., Signorile, J. F., Best, T. M., Jacobs, K. A., & Eltoukhy, M. (2023). A comparison of three-dimensional kinematics between markerless and marker-based motion capture in overground gait. *Journal of Biomechanics, 159*, 111793. doi:10.1016/j.jbiomech.2023.111793

Ripic, Z., Signorile, J. F., Best, T. M., Jacobs, K. A., Nienhuis, M., Whitelaw, C., . . . Eltoukhy, M. (2023). Validity of artificial intelligence-based markerless motion capture system for clinical gait analysis: Spatiotemporal results in healthy adults and adults with Parkinson's disease. *Journal of Biomechanics, 155*. doi:ARTN 11164510.1016/j.jbiomech.2023.111645

Ripic, Z., Signorile, J. F., Kuenze, C., & Eltoukhy, M. (2022). Concurrent validity of artificial intelligence-based markerless motion capture for over-ground gait analysis: A study of spatiotemporal parameters. *Journal of Biomechanics, 143*, 111278. doi:10.1016/j.jbiomech.2022.111278

Sabharwal, S., & Kumar, A. (2008). Methods for assessing leg length discrepancy. *Clinical Orthopaedics and Related Research, 466*, 2910-2922.

Sale, D. G. (1988). Neural adaptation to resistance training. *Medicine and Science in Sports and Exercise, 20*(5 Suppl), S135-145.

Schütz, P., List, R., Zemp, R., Schellenberg, F., Taylor, W. R., & Lorenzetti, S. (2014). Joint angles of the ankle, knee, and hip and loading conditions during split squats. *Journal of Applied Biomechanics, 30*(3), 373-380.

Shephard, R. J., & Shephard, R. J. (2015). The Renaissance: The Dawn of Scientific Enquiry into Health and Fitness. *An Illustrated History of Health and Fitness, from Pre-History to our Post-Modern World*, 347-445.

Shimokochi, Y., & Shultz, S. J. (2008). Mechanisms of noncontact anterior cruciate ligament injury. *Journal of Athletic Training, 43*(4), 396-408.

Slater, L. V., & Hart, J. M. (2017). Muscle activation patterns during different squat techniques. *Journal of Strength and Conditioning Research, 31*(3), 667-676.

Susruta, M. (2008). Susruta Samhita. *Ambika Dutta Shastri, editor, 2*, 39-41Susrutha.

Sutherland, D. H. (2002). The evolution of clinical gait analysis: Part II Kinematics. *Gait & Posture, 16*(2), 159-179.

Swinton, P. A., Lloyd, R., Keogh, J. W., Agouris, I., & Stewart, A. D. (2012). A biomechanical comparison of the traditional squat, powerlifting squat, and box squat. *The Journal of Strength & Conditioning Research, 26*(7), 1805-1816.

Tipton, C. (2008). Historical perspective: the antiquity of exercise, exercise physiology and the exercise prescription for health. In *Nutrition and Fitness: Cultural, Genetic and Metabolic Aspects* (Vol. 98, pp. 198-245): Karger Publishers.

Tipton, C. M. (2014). The history of "Exercise Is Medicine" in ancient civilizations. *Advances in Physiology Education, 38*(2), 109-117.

Van Den Bogert, A. J., Geijtenbeek, T., Even-Zohar, O., Steenbrink, F., & Hardin, E. C. (2013). A real-time system for biomechanical analysis of human movement and muscle function. *Medical & Biological Engineering & Computing, 51*, 1069-1077.

Van der Kruk, E., & Reijne, M. M. (2018). Accuracy of human motion capture systems for sport applications; state-of-the-art review. *European Journal of Sport Science, 18*(6), 806-819.

Van Grinsven, S., Van Cingel, R., Holla, C., & Van Loon, C. (2010). Evidence-based rehabilitation following anterior cruciate ligament reconstruction. *Knee Surgery, Sports Traumatology, Arthroscopy, 18*, 1128-1144.

van Schaik, J. E., & Dominici, N. (2020). Motion tracking in developmental research: Methods, considerations, and applications. *Progress in Brain Research, 254*, 89-111. doi:10.1016/bs.pbr.2020.06.007

Verhagen, E. A., van Tulder, M., van der Beek, A. J., Bouter, L. M., & van Mechelen, W. (2005). An economic evaluation of a proprioceptive balance board training programme for the prevention of ankle sprains in volleyball. *British Journal of Sports Medicine, 39*(2), 111-115. doi:10.1136/bjsm.2003.011031

Vincent, E., Max, R., Duncan Edward, M., Eline van, E., Ewoud van, A., Igor van den, B., . . . Marc, K. (2022). ACL reconstruction for all is not cost-effective after acute ACL rupture. *British Journal of Sports Medicine, 56*(1), 24. doi:10.1136/bjsports-2020-102564

von Staden, H. (1992). The discovery of the body: human dissection and its cultural contexts in ancient Greece. *Yale J Biol Med, 65*(3), 223-241. Retrieved from https://www.ncbi.nlm.nih.gov/pubmed/1285450

von Werder, S. C. F. A., & Disselhorst-Klug, C. (2016). The role of biceps brachii and brachioradialis for the control of elbow flexion and extension movements. *Journal of Electromyography and Kinesiology, 28*, 67-75. doi:https://doi.org/10.1016/j.jelekin.2016.03.004

Waldron, K., Brown, M., Calderon, A., & Feldman, M. (2022). Anterior Cruciate Ligament Rehabilitation and Return to Sport: How Fast Is Too Fast? *Arthroscopy, Sports Medicine, and Rehabilitation, 4*(1), e175-e179. doi:https://doi.org/10.1016/j.asmr.2021.10.027

Wang, N., Kozak, K., Wan, J., Gomez-Levi, G., & Strumolo, G. (2013). *Enhancing vehicle ingress/egress ergonomics with digital human models.* Paper presented at the Proceedings of the FISITA 2012 World Automotive Congress: Volume 7: Vehicle Design and Testing (I).

Warren, M., Lininger, M. R., Chimera, N. J., & Smith, C. A. (2018). Utility of FMS to understand injury incidence in sports: current perspectives. *Open Access Journal of Sports Medicine, 9*(null), 171-182. doi:10.2147/OAJSM.S149139

Weber, W. E., & Weber, E. (1836). *Mechanik der menschlichen Gehwerkzeuge: eine anatomisch-physiologische Untersuchung* (Vol. 1): in der Dieterichschen Buchhandlung.

Whittle, M. W. (1982). Calibration and performance of a 3-dimensional television system for kinematic analysis. *Journal of Biomechanics, 15*(3), 185-196. doi:10.1016/0021-9290(82)90251-2

Whittle, M. W. (1996). Clinical gait analysis: A review. *Human Movement Science, 15*(3), 369-387. doi:Doi 10.1016/0167-9457(96)00006-1

Wilk, K. E., Arrigo, C. A., Bagwell, M. S., & Finck, A. N. (2021). Considerations with open kinetic chain knee extension exercise following ACL reconstruction. *International Journal of Sports Physical Therapy, 16*(1), 282.

Wilk, K. E., Escamilla, R. F., Fleisig, G. S., Barrentine, S. W., Andrews, J. R., & Boyd, M. L. (1996). A comparison of tibiofemoral joint forces and electromyographic activit during open and closed kinetic chain exercises. *The American Journal of Sports Medicine, 24*(4), 518-527.

Yeadon, M., & Pain, M. (2023). Fifty years of performance-related sports biomechanics research. *Journal of Biomechanics*, 155, 111666. ISSN: 0021-9290.

Zahradka, N., Verma, K., Behboodi, A., Bodt, B., Wright, H., & Lee, S. C. K. (2020). An Evaluation of Three Kinematic Methods for Gait Event Detection Compared to the Kinetic-Based 'Gold Standard'. *Sensors, 20*(18), 5272. Retrieved from https://www.mdpi.com/1424-8220/20/18/5272

Zellmer, M., Kernozek, T. W., Gheidi, N., Hove, J., & Torry, M. (2019). Patellar tendon stress between two variations of the forward step lunge. *Journal of Sport and Health Science, 8*(3), 235-241. doi:https://doi.org/10.1016/j.jshs.2016.12.005

Zeni, J. A., Richards, J. G., & Higginson, J. S. (2008). Two simple methods for determining gait events during treadmill and overground walking using kinematic data. *Gait & Posture, 27*(4), 710-714. doi:https://doi.org/10.1016/j.gaitpost.2007.07.007